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High-performance self-compacting concrete with recycled coarse aggregate: Soft-computing analysis of compressive strength

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ABSTRACT

The growth of cities and industrialization has led to an increase in demand for concrete, resulting in resource depletion and environmental issues. Sustainable alternatives such as using recycled concrete aggregate (RCA) and industrial waste have been proposed to meet construction material demands while adhering to building codes and promoting sustainability. However, compressive strength (CS) is a crucial property of concrete, and the design parameters have different effects on CS for various grades. Recently, researchers have focused on partially replacing natural coarse aggregate (NCA) with RCA in concrete to achieve sustainability goals. This study aims to examine the influence of design parameters (w/c: water-cement ratio, w/b: water-binder ratio, A/c: total aggregate-cement ratio, FA/CA: fine-coarse aggregate ratio, SP: superplasticizer, w/s: water-solid ratio and RCA%) on concrete CS and address controversies in the insights gained from pairwise comparisons using Pearson's correlation coefficient (PCC) analysis. Additionally, five techniques (M5P, RF, SVM, LR, and ANNs) were used to predict the CS of high-performance self-compacting concrete (HP-SCC) with RCA, and the results were compared with an ANNs-based model as was the commonly used one in literature. The approaches were assessed based on their accuracy measured using correlation coefficient (CC), mean absolute error (MAE), Root Mean Square Error (RMSE), Mean absolute percentage error (MAPE), Scatter index (SI), and comprehensive measure (COM) indicators. Accordingly, the analysis indicated that SVM-PUK-based model is the most appropriate and effective technique to predict the CS of HP-SCC for the given datasets, with CC = 0.894, 0.900, MAE = 1.721, 3.813, RMSE = 5.137, 6.306, and MAPE = 4.5%, 7.6% for the training and testing stages, respectively. The uncertainty analysis results were 21%, 20.7%, 19%, 22%, and 19% for M5P, RF, SVM, LR, and ANN-based models, respectively, whereby all of them were under threshold of 35%. Moreover, according to sensitivity analysis, w/c, w/b, and w/s variables influences the most on CS prediction, while the RCA(%) variable has the least impact.

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1. Introduction

Construction and demolition waste (CDW) refers to waste materials that are generated from construction, renovation, or demolition activities. This type of waste used to be just thrown away, but now there is the establishment of waste legislation in the form of guidelines and directives in many countries that say this type of waste must be managed eco-friendly. Thus, companies in the construction industry are working on reducing the amount of waste produced by recycling or reusing as much CDW as possible [1,2]. The building sector is a significant material consumer and waste generator [3]. Concrete constitutes 70% of CDW, according to Abbas et al. (2009) [4].

Reasons for reusing CDW preserving limited materials are:

- CDW occupies a significant amount of landfill space. If CDW is not reused, it will continue to take up valuable space that could be used for other purposes.
- CDW contains various materials like wood, metal, and concrete which can all be recycled and used in new construction projects.

Table 1
Studies conducted on soft computing techniques for the CS prediction of SCC.

The used soft computing techniques and the range of CS for modelling

Previous works	Predictive model(s)	Range of CS MPa
[19]	ANN	2.04–122
[20]	ANN	9–40
	MR	
[12]	LR	52–77
	ANN	
	ANFISs	
[11]	ANN	33–46
	M5P	
	RF	
[30]	LR	16–33
	NLR	
	RF	
[13]	M5P	33–70
	RT	
	SVM	
[17]	ANN	1.1–113.1
	ANN	
	ANFISs	
	M5P	
	SVM	
[14]	ANN	2.33–82.6
	SVM	
[15]	ANN	9.89–105.7
	LR	
	NLR	
	MLR	

Proportional utilization of soft computing techniques in a cohort of 65 peer-reviewed journal papers derived from reliable Publishers, encompassing Elsevier, Springer, ACI, and ASCE [31].

Technique	Percentage of use	Advantages
ANN	43.9%	• "Able to predict accurately even when working with poor data" • "Self-adapting model to different environmental conditions" • "Parallel processing capabilities" • "Due to its generalization capability, ANN models can predict accurate results of experiments other than the ones it was trained for"
Statistical Procedures	12.3%	-
SVM	11.4%	• "Can overcome the problem of small sample size" • "Always identifies a global minimum and not a local one" • "Requires smaller computational time than ANN models"
Tree-based Models	10.5%	• "Merging predictor categories help to avoid overfitting"
Genetic Methods	9.6%	• "Adapts to changing environments • Ability to handle various types of objective functions (root mean square error, sum squared error, etc.)" • "Overcomes the disadvantage of the black box algorithms"
Others	7.0%	-
FL	5.3%	• "A powerful tool to simulate non-linear behavior"

Note: R^2 is the correlation coefficient and FL is Fuzzy Logic.

- CDW is a significant source of pollution. If this waste is not reused, it will release harmful chemicals and pollutants into the environment.

If this waste is not reused, it will release harmful chemicals and pollutants into the environment.

In response to these issues, the concept of making sustainable concrete — by definition, concrete made from recycled aggregate — has evolved to qualify as a "sustainable" material; recycled concrete aggregate (RCA) must:

- be able to recycle CDW and minimize the use of natural resources.
- be able to have negligible negative environmental impacts.
- be able to contribute to maintaining ecological sustainability and development.

RCA combines old concrete fragments with new cementitious matrix, and technological challenges constrain the quality of RCA. It is important to remember that each RCA particle is still made up of natural coarse aggregate (NCA) and attached adhered mortar (AM) around them. The separate identification of concrete components is required for thorough comprehension and standards of the RCA and for making predictions about its potential impacts on concrete because each component has different properties that can affect the overall performance of the concrete. For example, the type of cement used, the amount of water used, and the aggregate used can affect the concrete's strength, workability, and durability [5]. The qualities of the RCA, the mix design, the mixing method, and the state of deterioration of the RCA contribute to the final product's quality. The mechanical attributes of RCA are often the emphasis of all research on the usage of RCA. It is an adopted concept that although it is conceivable to use RCA as an NCA, using it will often cause a reduction in RCA strength. This is a circumstance that could be handled by raising the amount of cement, changing the water-cement ratio, adding supplementary cementitious materials (SCMs) to the mixture, etc. [6].

Concrete is a versatile building material used for construction over the last few decades. It is made by mixing cement, water, and other materials and then compacting it to make it stronger. Self-compacting concrete (SCC) is a type of concrete that is easier to work with because it does not require compaction as much. It can be used for more complicated projects than regular concrete. High-performance Self-compacting concrete (HP-SCC) is an SCC with high compressive strength (CS) [7]. Generally, such an HP-SCC is obtained with high cement ranges and a low water/cement (w/c) ratio, which can be improved by using inactive or active supplementary materials to the mix [8]. To ensure that materials will last a long time and be sustainable, various design parameters have been utilized in previous studies on the mechanical characteristics of HP-SCC. This updated work allows for a compendium of the advances accomplished so far in this knowledge area that will help establish the design criteria where there needs to be a clearer understanding of correlations between the design parameters of the hardened concrete properties made incorporating up to 100% RCA and SCMs.

The engineers' difficulties in predicting the CS of HP-SCC with RCA and SCMs are: 1) The RCA may have a different composition than the original concrete aggregate, which can affect the strength of the concrete [9]. 2) The RCA may be of a different size than the original concrete aggregate, which can also affect the strength of the concrete [9]. 3) The RCA may have been contaminated with other materials, which can also affect the strength of the concrete [10]. A computational model can efficiently predict the strength of concrete because it can take into account the effects of the different ingredients on the concrete's strength. Many soft computing methods have been used to predict the CS of concrete in the last years, including decision tree (M5P) [11], random forest (RF) support vector machines (SVM) [12–14], linear regression (LR), nonlinear regression (NLR) [15], multiple linear regression (MLR) [15] adaptive network-based inference systems (ANFISs) [16,17], and artificial neural networks (ANNs) [18–20], where the soft computing model of concrete strength prediction is a model that uses artificial intelligence (AI) techniques to predict the strength of concrete based on its variables. Nevertheless, no study has conducted computing analysis for only HP-SCC (CS>40 MPa) except for some studies limited to implementing LR, NLR, MLR ANN, and ANFISs. In contrast, most researchers conducted the analysis using cross-strength datasets (refer to Table 1). In a related context, the mechanism of the soft computing model for CS prediction is based on several factors, including the material's composition (type of cement used, the amount of water used, the aggregate used), the curing process, the age of the concrete, and the environmental conditions [21]. Therefore, as long as the design parameters of the concrete mixes are concerned with producing high-quality concrete, detailed data from the experiments should be obtained. A data-driven analysis should be conducted to identify all pairs of concrete design parameters in the datasets sourced from HP-SCC literature [22,23]. In addition, it provides in-depth and cross-categorical perspectives of the relationships between each variable pair and scientifically grounded perspicuity about the experimental results that might be used to estimate other variables.

In light of the preceding, the advantages of soft computing models include their ability to deal with imprecision, uncertainty, and partial truth; their tolerance for noise and error; and their capacity for learning from data [24]. Additionally, soft computing models are often more efficient and less expensive to develop and implement than traditional models [25]. On the other hand, there are a few disadvantages to the soft computing model, which include: 1. It can be challenging to implement, as it requires a lot of data and computing power [26]. 2. It can be slow, as it takes time to process all the data [27]. 3. It can be inaccurate, as it relies on probabilistic methods [28]. It is worth noting, in recent years, neural networks have seen a resurgence due to the use of ANNs in many scientific fields. Therefore, ANNs are often used to predict the behaviour of structural materials and assess their CS, as shown in Table 1. Whereas, for a small number of datasets, SVM is recommended which requires less computational effort than the ANNs model [29].

Since implementing RCA in HP-SCC can potentially affect its CS, which is an important property, the assessment of CS using computational techniques such as M5P, RF, SVM, LR, and ANNs is required. These techniques utilize statistical and mathematical models to predict the CS based on design parameters. Using these computational techniques, it is possible to accurately predict the CS, which can help optimize the mix design and ensure the structural integrity of the concrete.

The main objective of the study is to develop computational models for predicting the CS of sustainable HP-SCC with RCA using various mixing design parameters. The authors conducted this computing analysis and investigated the influence of design parameters on the CS using extracted datasets from the literature, including 111 experimental results. First, a correlation analysis of datasets was developed using *Python version 3.7.15* to understand the relationship between each pair of variables; then second, five soft-computing techniques (M5P, RF, SVM, LR, and ANNs) were implemented. Prediction models of the CS were compared through assigned indicators, including correlation coefficient (CC), mean absolute error (MAE), Root Mean Square Error (RMSE), Mean absolute percentage error (MAPE), Scatter index (SI), and comprehensive measure (COM) to evaluate and select the best prediction model as well as rank all the other models accordingly.

The primary contribution of the work is to:

- Add new design parameters to the literature for HP-SCC containing RCA.
- Understand the relationship between all the variables through statistical procedures such as Pearson's correlation coefficient (PCC) and Variance inflation factor (VIF)
- Provide models that can accurately predict the CS of sustainable HP-SCC with RCA, which the construction sector can adopt for real-life applications.
- Present empirical equations with simplified forms and superior accuracy compared to the existing ones in the literature.
- Identify the most influencing parameters on the CS.

The present study will help to reduce the need for laboratory experimentation and analytical impediments, thereby saving time and costs for the construction sector.

Section 2 reviews machine learning tools of the computing analysis. Section 3 presents the framework, materials, and methods. Section 4 discusses primary outcomes regarding the application of methods and the practical insights gained from the work. Section 5 presents the uncertainty analysis, Section 6 shows the conducted sensitivity analysis, Section 7 insights into the limitations of the models. Finally, Section 8 provides conclusion of this paper.

2. Machine learning tools

2.1. M5P-tree model (M5P)

M5P-tree is one of the genetic algorithms first introduced by looking at regression problems [32]. The M5P-tree method is based on continuous class issues rather than discrete segments, and it can solve functions with very high dimensionality [33] (see Fig. 1). The established data of each model's variable is revealed to estimate the nonlinear correlation between datasets. For each node on the M5P-tree model tree division, error estimations and parameter information are provided. The node's class and input default values are checked for problems [34]. The attributes capable of maximising the predicted error are used to evaluate each node's feature. The M5P tree model is based on nodal error estimation. After knowing the division parameters of the M5P-tree model, the tree is collected based on nodal error predictions [35]. The M5P defect is evaluated using the standard class value variance at the node [36]. The separation frequently results in a complex arboreal system, which adds to over-structure. In the second stage, the giant trees are pruned, and linear regression functions replace the trimming sub-trees [33,34,37–41].

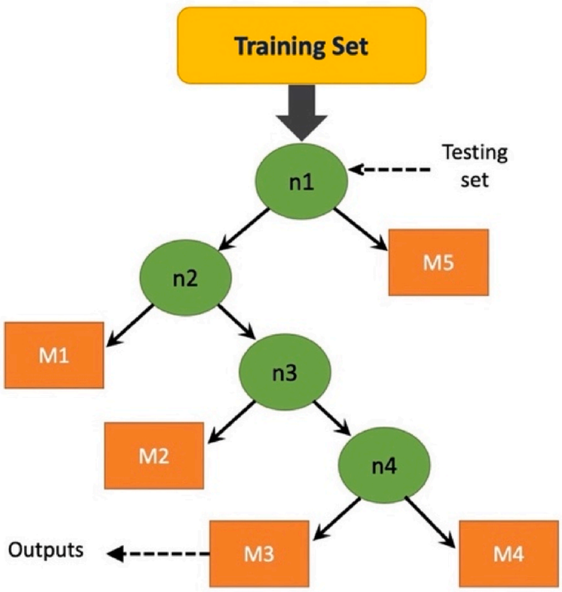

Fig. 1. Flowchart of M5P-tree.

2.2. Random forest (RF)

RF is a powerful ensemble learning method that uses a combination of decision trees and bootstrap aggregating (bagging) to improve the accuracy and generalization of the model [42]. The RF algorithm creates multiple decision tree models by randomly selecting a subset of the input features and training each tree on a different training data sample as outlined in Fig. 2 using bootstrap aggregating [43]. RF is an ensemble learning method created by a decision tree model {h(X, θk), k = 1, ⋅⋅⋅ K} which uses bagging integration [44], where k is an independent random vector with the same distribution [44]. Input sample x into RF and find the output data f(x) = majority {h(X, θk), k = 1, ⋅⋅⋅ K} [45]. RF is capable of accurately classifying a wide range of data. Once the tree models are trained, RF combines their predictions using a majority voting scheme for classification or an average for regression problems [46]. This helps to improve the overall prediction's accuracy and reliability and provides a measure of uncertainty for each prediction. RF is a versatile algorithm that can handle various types of data and has been widely used in many applications, such as the mechanical properties of concrete. However, it is essential to note that RF may not always be the best algorithm for every problem, and it is important to compare its performance with other models and choose the one that works best for the specific task. It can pross a wide variety of input parameters [11,34,39,40]. The most significant disadvantage of RF is that many trees will be generated, which hinders real-time efficiency. The RF algorithm is usually the simplest to train but making calculations after practice requires time. Increased reliability requires more trees and performance. Although the random forest algorithm is fast enough for most real-world implementations, runtime efficiency might be required in some cases.

2.3. Support vector machine (SVM)

Vapnik proposed in 2000 [47] that SVMs are a machine learning algorithm based on statistical principles. SVMs have been considered a novel approach for classifying problems and are often recommended as suitable for such situations. By adding Vapnick's insensitive loss function, SVM for regression (SVR) was developed to handle non-linear regression estimation problems [48,49]. SVR has expanded its application beyond its original domain and is currently being utilized in various fields such as optimal controls, time series forecasting, and regression interval analysis [48,50,51]. Notwithstanding, selecting several parameters, referred to as hyper-parameters, is a critical aspect of SVM or SVR [34–36,52,53]. The epsilon value proposed by Vapnik encompasses the kernel function, the insensitive loss function, and a continuous regularization term (referred to as the complexity parameter C). There are several different kernels, but the polynomial and Gaussian radial database kernels (RBF) are two kernels that are widely used for real-world data. The degree polynomial kernel (d) shall be described as (Eq. (1))

$$K(x_i, x_j) = (x_i \bullet x_j + k)^d \tag{1}$$

Where the constant is k. The function becomes a linear one when the kernel with d is equal to 1. The second used one is the Gaussian radial base function (RBF) kernel described by the kernel (Eq. (2)).

$$K(x_i, x_j) = exp\left(-\frac{\|x_i \bullet x_j\|}{2\gamma}\right)^2 \tag{2}$$

Where $\gamma > 0$ is the parameter that holds the Gaussian distance. It plays a function close to the polynomial kernel's degree in regulating the versatility of the subsequent classifier.

The other kernel functions SVM such as the Pearson VII Universal Kernel (PUK) [54]. The following relationship (Eq. (3)) gives the overall shape of PUK for curve-fitting purposes:

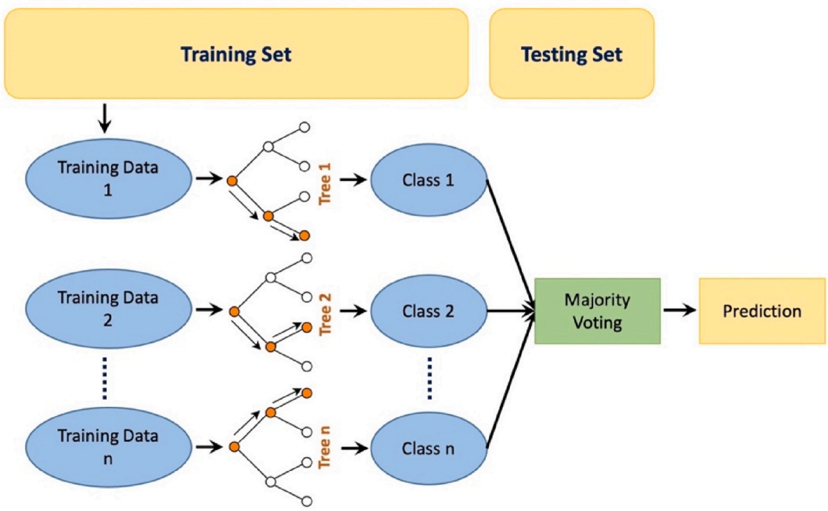

Fig. 2. Flowchart of RF.

$$f(x) = H \Bigg/ \left[1 + \left(\frac{2(x - x_0)\sqrt{2^{\frac{1}{w}} - 1}}{\sigma}\right)^2\right]^w \tag{3}$$

The peak height is called H at the peak's middle x_0, and even x reflects independent variables. All parameters, ω, and σ control the half-width, also known as the Pearson width and peak tailing factor, are regulated by all parameters [55]. The primary reason for using the Pearson VII curve fitting function is to change the parameters w and σ. The following formula (Eq. (4)) defines the Pearson VII multi-dimensional input space kernel function:

$$K(x_i, x_j) = 1 \Bigg/ \left[1 + \left(\frac{2\sqrt{\|x_i \bullet x_j\|^2}\sqrt{2^{\frac{1}{w}} - 1}}{\sigma}\right)^2\right]^w \tag{4}$$

2.4. Linear regression (LR)

One of the most frequently utilized regression equations for estimating the CS of concrete (Eq. (5)) is the LR model [56] (refer to Fig. 3).

$$CS = a + b(w / b) \tag{5}$$

Where w/b is the water ratio, and b is the model's parameters. Due to this, both show the reflectivity of concrete (MPa) and the equation for the parameters. However, concrete mortar is not included in the calculation because it is a variable affecting CS, such as the curing time of the cementitious material [36]. It is suggested that all other variables should be included to obtain more accurate performance. (Eq. (6)).

$$f_{st} = a + b\left(\frac{\boldsymbol{w}}{\boldsymbol{c}}\right) + c * \left(\frac{\boldsymbol{w}}{\boldsymbol{b}}\right) + d * \left(\frac{\boldsymbol{w}}{\boldsymbol{s}}\right) + e * \left(\frac{\boldsymbol{A}}{\boldsymbol{c}}\right) + f * \left(\frac{\boldsymbol{FA}}{\boldsymbol{CA}}\right) + g * (\boldsymbol{SP}\ (\boldsymbol{kg})) + h * (\boldsymbol{RCA}\ (\%)) \tag{6}$$

Where a, b, c, and d are the model's parameters. Eq. (5) predicts that all variables are linear in extent, as can be shown in Eq. (6). Even yet, it may not have occurred in all instances owing to factors such as cement mortar mix variations and their interrelationship influencing both strength and fracture. The improvement described here may not have occurred in all cases. As a result, models must be modified regularly to predict the CS of concrete with an appropriate level of precision [34].

2.5. Artificial neural networks (ANNs)

ANNs are a computational model that is similar to how the human brain processes information. They are also machine learning systems that can be used for various numerical estimations and challenges in Civil Engineering [57]. Three layers comprise an ANN system: an input layer, a hidden layer (or layers), and an output layer as shown in Fig. 4. Weight, transfer function, and bias connect the hidden layer with the other layers [41,58–61]. Proportions include inputs, and outputs were programmed into a multi-layer feed--forward network. No accepted process exists for deciding on or creating a network architecture. Through the application of a trial-and-error approach utilizing the criterion of minimal Mean Absolute Error (MAE), the optimal number of hidden layers and neurons was determined [62]. Identifying an appropriate number of training epochs that yields low MAE, RMSE, and CC constitutes the second stage in developing an optimal neural network [63]. Neural networks utilizing hyperbolic tangent transfer functions were utilized to investigate the effects of varying the number of epochs on the reduction of MAE and RMSE. The study involved the analysis of MAE changes as a function of epoch count for the initial neural network models.

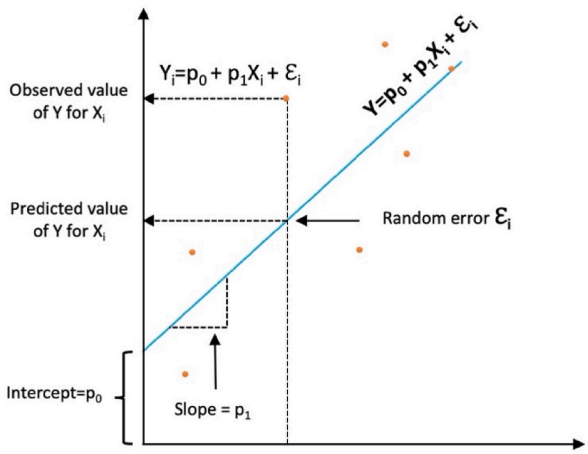

Fig. 3. Flowchart of LR.

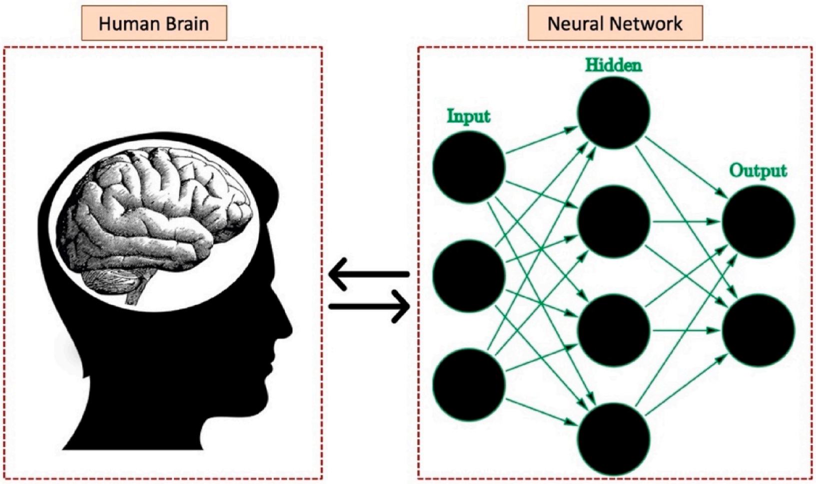

Fig. 4. Flowchart of ANNs.

3. Materials and methods

In this work, 15 studies from the literature have been studied for extracting the dataset and six computational approaches have been implemented to assets them.

3.1. Details of the dataset

The reviewing methodology followed for this study is described in Fig. 5. The data collection strategy consisted of collecting publications about HP-SCC while rejecting all unrelated articles. SCC articles with RCA are prioritized, and the remaining articles are eliminated to provide an in-depth study for a better understanding of the development mechanism. The SCC mixture's essential characteristics could be enhanced by using additives where not only the price of the combination can be minimized. Articles that include admixtures are ultimately highly valued and given more emphasis. The design parameters were derived using new different mix designs for evaluating CS of the literature where the keywords used in the search are: "High-strength self-compacting concretes", "Properties of Self-consolidating Concrete", "self-compacting concretes made with Recycled Aggregates", "Influence of Recycled Aggregates on Properties of Self-consolidating Concretes", "Construction and Demolition Waste in Self-Compacting Concrete", "Performance of Self-Compacting Concrete Made with Coarse Recycled Concrete Aggregates", "Self-compacting concrete for environmental sustainability", "Development of High-Performance Self-Compacting Concrete Using Waste Recycled Concrete Aggregates", "Properties of Concrete Prepared with Coarse Recycled Aggregate", and "compressive strength of self-compacting concrete",

Mix design refers to determining the relative amounts of individual ingredients in kilograms per cubic meter, which allows SCC to achieve the desired fresh and hardened concrete characteristics, but here it has been chosen to be done by ratios for easier recognition. Based on 111 published mix designs, the design parameters were obtained from Q1 journals for more reliability. In this analysis, researchers consider the following design parameters: w/c: water-cement ratio, w/b: water-binder ratio, A/c: total aggregate-cement

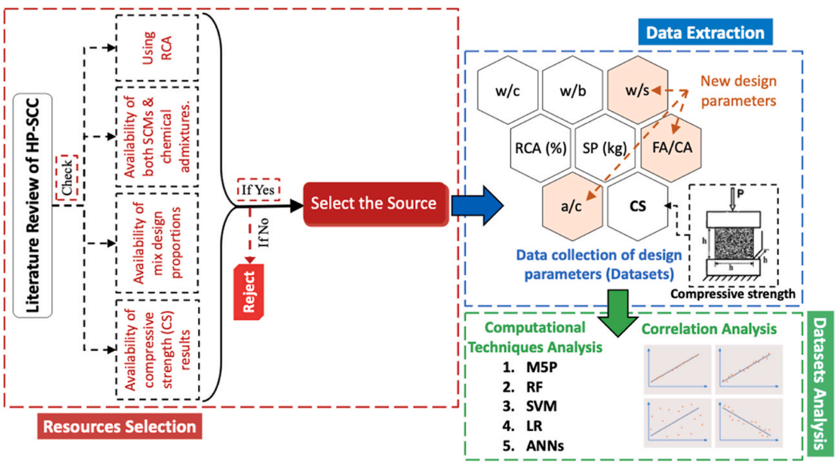

Fig. 5. Methodology of reviewing the literature, extracting the dataset, and conducting a soft-computing analysis.

ratio, FA/CA: fine-coarse aggregate ratio, SP: Superplasticizer, w/s: water-solid ratio and RCA% as shown in Fig. 5. In mixing proportions, the design parameters were extracted based on the availability of components. Studies with less variety of substances were not considered for the analysis. Also, based on CS grade, the current study classified the data into HP-SCC datasets with a 28-day CS 40 MPa–80 MPa range regardless of the code used for testing.

3.2. Design parameters in the literature

Table A in Appendix displays 15 studies from the literature with CS between 40 and 80 MPa at 28 days and shows the various mixture proportions used in each production. Table A of Appendix A shows that a lower w/c or w/b ratio is associated with greater CS. The influence of concrete variables on various CS grades of HP-SCC concrete with RCA is illustrated in Fig. A (refer to Appendix A), all according to the literature from 2010 to 2020. For all mixtures, parameters vary as A/c ratio (3.31 to 12.4), FA/CA ratio (1.51 and 6.04), and SP (1.2 to 11 kg), while RCA content of the mixes varies between 0 and 100%. Table 2 shows the statistical description of inputs and output variables from these extracted literature articles published between 2010 and 2020 of HP-SCC with RCA. Worth to mention, some independent variables were heavily skewed, such as A/c and FA/CA. Thus, it is important to normalize them before the analysis using filter [64]. The graphical representation histograms of datasets are shown in Fig. 6. (Histogram of the Inputs) and Fig. 7. (Statistical analysis of the output).

3.3. Pearson's correlation coefficient (PCC)

PCC (r) is a standard method for estimating a linear correlation. A value lies between –1 and 1 and measures the CS and direction of the association between two variables [65,66]. Here the PCC method is the basis of the analysis used to identify the correlation between each pair of design parameters (variables).

The PCC is typically defined and computed as (Eq. (7)) (Eq. (8)) (Eq. (9)) as the following

$$r = \frac{covariance\ (x, y)}{\sqrt{var(x)}\sqrt{var(y)}} \tag{7}$$

Where:

$$covariance(x, y) = \frac{\sum_{i=1}^{n}(x_i - \overline{X})(y_i - \overline{Y})}{n - 1} \tag{8}$$

$$var(x) = \frac{\sum_{i=1}^{n}(x_i - \overline{X})^2}{n - 1} \tag{9}$$

$\overline{X}, \overline{Y}$ are the mean values of the compared data and 'n' is the number of samples in the data.

There are two primary reasons for using the PCC method [66]:

The first reason for using the PCC is that it is specifically designed for use with continuous variables. Other variables, such as nominal or ordinal variables, would require different measures of association. The second reason for using the PCC is that it has been widely used and studied for over a century, making it a well-established and widely recognized measure of correlation. It is also easy to compute and interpret, making it a popular choice for many applications. However, it is essential to note that the PCC assumes a linear relationship between the variables and may not be appropriate for variables with a nonlinear relationship. Additionally, the PCC measures only the strength of a linear relationship and does not provide information on the shape or form of the relationship.

Table 2
Statistical analysis of variables.

	Inputs							Outputs
	w/c	w/b	A/c	FA/CA	SP (kg)	w/s	RCA (%)	CS (MPa)
Mean	0.432	0.295	4.564	2.332	4.100	7.599	45.757	48.558
Standard Error	0.009	0.007	0.161	0.078	0.230	0.224	3.908	1.178
Median	0.430	0.310	4.010	2.148	3.890	8.070	40	46.540
Mode	0.400	0.320	3.488	1.884	1.230	11.630	0	42.340
Standard Deviation	0.095	0.079	1.693	0.819	2.419	2.363	41.174	12.410
Sample Variance	0.009	0.006	2.867	0.671	5.850	5.585	1695.331	154.002
Kurtosis	1.947	0.123	10.900	9.598	-0.395	-0.256	-1.560	0.455
Skewness	0.806	-0.778	2.943	2.706	0.546	-0.252	0.259	0.321
Range	0.580	0.320	9.328	4.528	9.640	9.500	100	67.760
Minimum	0.250	0.090	3.110	1.510	1.160	2.130	0	13.640
Maximum	0.830	0.410	12.438	6.038	10.800	11.630	100	81.400
Count	111	111	111	111	111	111	111	111
Confidence Level (95.0%)	0.018	0.015	0.318	0.154	0.455	0.445	7.745	2.334

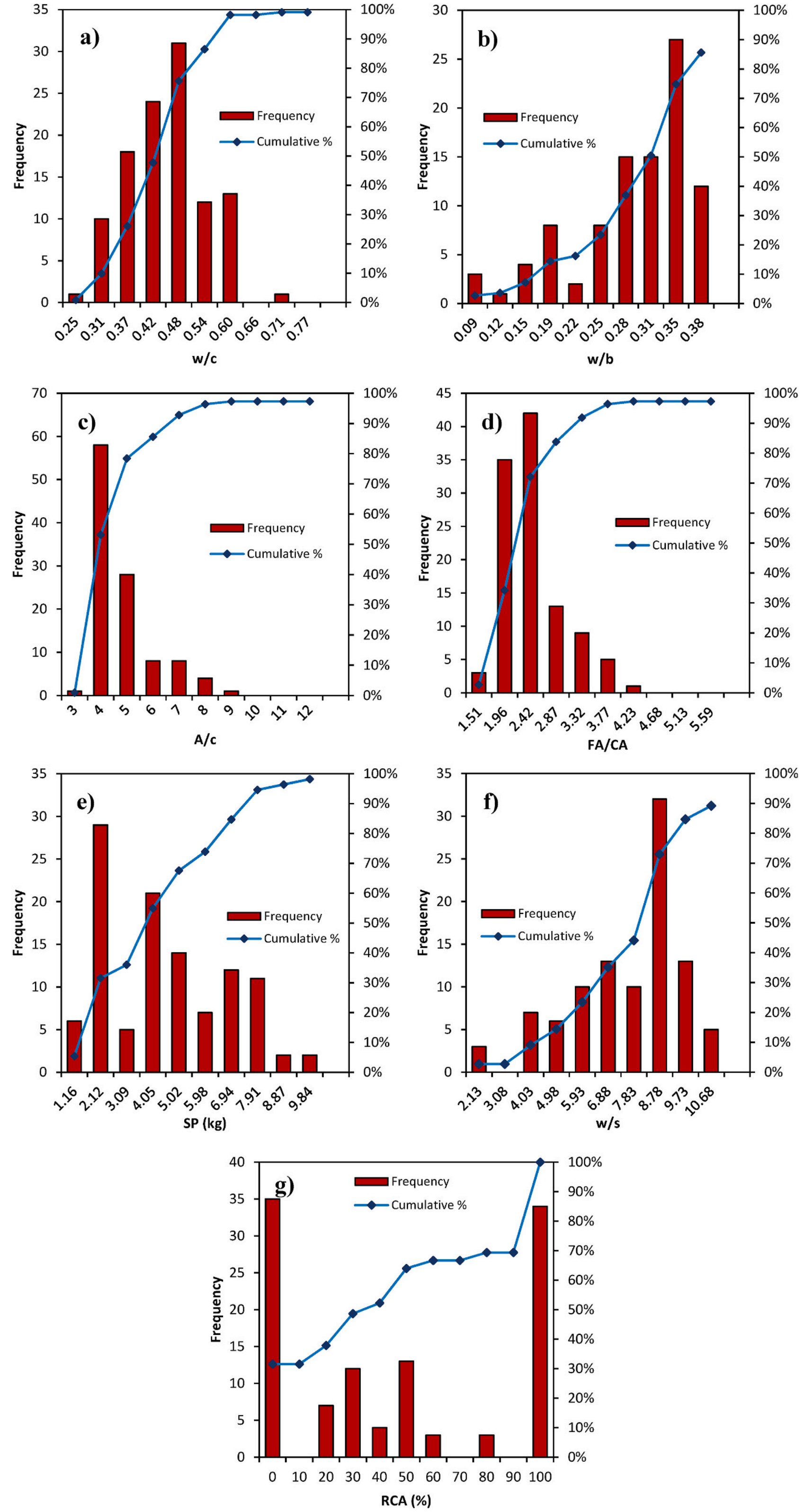

Fig. 6. Histograms of the input variables used in this study: (a) w/c; (b) w/b; (c) A/c; (d) FA/CA; (e) SP (kg); (f) w/s; (g) RCA%.

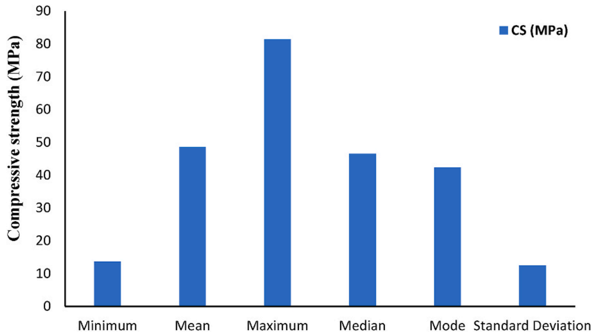

Fig. 7. Statistical analysis of the output (CS) variable.

3.4. Multicollinearity

Multicollinearity arises when there are substantial intercorrelations among two or more predictor variables within a multiple regression framework. This phenomenon can introduce distortions or misrepresentations when researchers or analysts strive to ascertain the optimal utilization of each independent variable for predicting or comprehending the dependent variable within a statistical model. Moreover, it is not enough just to look at the CC (section 3.3) only because CC gives an isolated view of the effective input variable on the output which does not account for the overlap with all of the other variables. In other words, for understanding the multilinearity in terms of how much the independent variable is correlated not just with only one variable but with all the other variables at the same time. Thus, the variance inflation factor (VIF) is important to be calculated.

In the realm of statistical analysis, VIF holds a pivotal role as a diagnostic tool for evaluating the presence of collinearity among independent variables [67]. The VIF value can be calculated using Eq (10),

$$VIF_i = \frac{1}{\left(1 - R_i^2\right)}, i = 1, \ldots, k \tag{10}$$

Where:

k is the number of independent variables;
R_i^2 is the coefficient of determination obtained from a fitted linear regression model on the ith variable as the target with the other independent variables.

VIF numerical range spans from 0 to 10, occasionally limited to 0 to 5, providing a quantifiable measure of the extent of collinearity within a regression model [68]. Additionally, the tolerance value, which corresponds to the reciprocal of the VIF, aids in the assessment of multicollinearity. A tolerance exceeding 0.1 but remaining below 1 is conventionally considered an indication of the absence of significant multicollinearity [69]. Consequently, as displayed in Table 2, the absence of multicollinearity is evident among the independent variables, given that the VIF values are below 5 and the tolerance values surpass 0.1 [69].

3.5. Assessment criteria of models performance

The efficiency of the modelling was estimated using the abovementioned CC, MAE, RMSE, MAPE, and SI values [34,42,70]. CC, MAE, RMSE, MAPE, and SI are widely used statistical performance assessment indicators to assess the efficiency of the computational approaches for a distribution of 70% training dataset and 30% testing dataset. These indicators complement each other in assessing model performance. Therefore, the reason behind using this metric solely is:

- Commonly used and aligned with the literature
- Suitable for the data and balanced between simplicity and comprehensiveness

It is worth noting that while the selection of specific metrics is important, it is also valuable to report multiple metrics to present a more comprehensive picture of the model's performance. This helps to address the limitations of individual metrics and provides a more nuanced evaluation. In the implemented metrics higher CC with petite MAE, RMSE, MAPE, and SI indicate better model accuracy, the metrics are calculated as follows (Eq.(11)–(15)).

$$CC = \frac{\sum_{i=1}^{n}(O_i - \overline{O})(P_i - \overline{P})}{\sqrt{\sum_{i=1}^{n}(O_i - \overline{O})^2}\sqrt{\sum_{i=1}^{n}(P_i - \overline{P})^2}} \tag{11}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|P_i - O_i| \tag{12}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{n}(P_i - O_i)^2} \tag{13}$$

$$MAPE = \frac{100}{N}\sum_{i=1}^{n}\frac{|O_i - P_i|}{O_i} \tag{14}$$

$$SI = \frac{RMSE}{\bar{x}} \tag{15}$$

Where: $O_i, P_i, n, and\ \bar{x}$ are Actual value, predicted value, number of data, and the mean actual data, respectively.

Frankly, the evaluation outcomes of these aforementioned metrics may not always align, thereby necessitating the need for a holistic measure. In this regard, a comprehensive measure (COM) has been proposed, serving as a unified approach to assess model performance. The COM can be calculated by considering all three metrics, offering a more encompassing evaluation criterion. By incorporating RMSE, MAPE, and CC into a single measure, the COM accounts for the potential inconsistencies among individual metrics and provides a more comprehensive assessment of a model's overall performance [71], COM can be calculated as follows (Eq. (16)).

$$COM = \left(\frac{1}{3}\frac{RMSE_{Training} * MAPE_{Training}}{CC_{Training}}\right) + \left(\frac{2}{3}\frac{RMSE_{Testing} * MAPE_{TTesting}}{CC_{Testing}}\right) \tag{16}$$

The allocation of weight to performance on the training and testing sets was determined to be 1/3 and 2/3, respectively. This decision was made based on the understanding that evaluating the model's performance on the testing set provides insights into its generalizability, and thus deserves greater consideration. It is important to note that a lower COM value indicates a superior overall modeling performance [71].

Note: More metrics will be used in the upcoming sections (for uncertainty analysis).

4. Result and discussion

4.1. Correlation analysis

An analysis of the correlation between the input and output variables was conducted to establish a relationship between the datasets, and this statistical investigation informed the design of a predictive model that ultimately enhanced the accuracy of the predictions. So for this reason, the correlation between the independent inputs using *Python version 3.7.15 was* analysed, as shown in Fig. 8 using the heat map. When there is a correlation ($|r| > 0.5$) between the input variables, it may suggest the presence of multicollinearity, which can affect the modeling results and potentially introduce bias into the model. As seen in the heat map, w/b & w/s (r = 0.9), FA/CA & A/c (r = 0.84), w/c & w/s (r = 0.72), and w/b & w/c (r = 0.72), respectively greater than 0.5, which lead to understanding that these inputs have a tremendous strong positive relationship between them although there was a substantial negative connection between some of the other inputs, such as between w/s & A/c (r = −0.72), w/b & A/c (r = −0.67), w/c & CS (r = −0.5), and FA/CA & w/s (r = −0.46). The rest is just average. As long as some of the inputs correlated $|r|$ greater than 0.5, showing that multicollinearity did happen. It is worth mentioning that in w/b & w/s and FA/CA & A/c, r was >0.8, indicating multicollinearity between them. thus, there is a need for the Variance inflation factor (VIF) assessment (which will be discussed in the next section 4.2.).

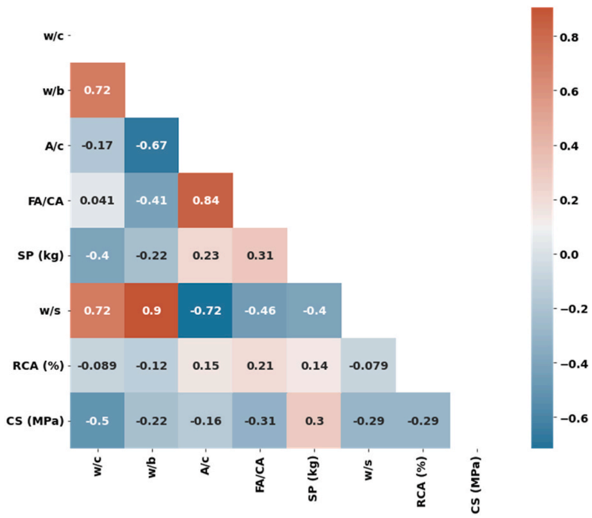

Fig. 8. Multi-correlation between the variables through a heat map.

4.2. Variance inflation factor (VIF)

Table 3 presents the calculated VIF and the tolerance values. VIF and tolerance values of all the independent input variables are less than 5 and greater than 0.1, respectively, which reflect the observation that no found collinearity among the inputs that might lead to distort the prediction models. However, the highest VIF and lowest tolerance might be the reason behind being w/c and w/b are the most important variables of the sensitivity analysis in the literature [72–78].

4.3. M5P model predictions

Building a model with M5P involved trial and error, as different model structures were tested to see which one performed best on the data, and the structures are shown in Fig. 9. Table 4 presents established linear equations using a trimmed M5P-based model. Once a model has been built, it was evaluated using metrics such as the CC, RMSE, and MAE. In the case of the M5P concrete model, the model performed well on both the training and testing datasets, with a good CC. Fig. 9 shows the relationship between the actual and predicted CS of HP-SCC for both the training and testing dataset stages. As mentioned in Table 5, the M5P-based model performs well in the prediction with CC = 0.801, 0.791, MAE = 4.900, 7.099, RMSE = 6.860, 8.799, and MAPE = 10.7%, 13.8% for training and testing phases, respectively, which suggests that the model was able to accurately predict the CS of HP-SCC based on the input data.

4.4. RF model predictions

Developing an RF model is similar to developing an M5P model in that it involves training multiple decision trees on the data and then combining their predictions to make a final one. Fig. 10 shows the agreement plot between actual and predicted concrete CS of HP-SCC for the training and testing datasets. The prediction values of the RF-based model show well performance in both stages (training and testing). The RF-based model was able to accurately predict the CS of HP-SCC based on the input data with CC = 0.966, 0.887, MAE = 2.212, 5.044, RMSE = 3.230, 6.951, and MAPE = 4.8%, 9.8% for the training and testing stages, respectively (Refer to Table 5). The accuracy was higher and better than the M5P-based model owing to MAPE < 10% in both stages. Frankly, the RF-based model works by training multiple decision trees on different subsets of the data and then combining the predictions made by each tree to make a final one. Thus, this always helps improve performance and reduce overfitting with fewer errors.

4.5. SVM model predictions

SVM is a supervised algorithm that can be utilized for both classification and regression tasks. In the case of regression, SVM can identify a hyperplane that separates the data into different classes or predicts continuous outcomes. Two kernel functions, PUK and RBF, were employed to develop input SVM models. Results indicated that the SVM-PUK-based model outperformed the SVM-RBF-based model, as demonstrated by higher values of performance metrics such as CC, RMSE, and MAE. Fig. 11 illustrates the high relationship between actual and predicted values of CS of HP-SCC. Although, the model slightly overestimated some of the low-strength concrete mixes. Furthermore, the performance of the SVM-PUK-based model was superior, with CC = 0.894, 0.900, MAE = 1.721, 3.813, RMSE = 5.137, 6.306, and MAPE = 4.5%, 7.6% for the training and testing stages, respectively (refer to Table 5). These findings suggest that the SVM-PUK-based model can accurately predict the CS of HP-SCC in both the training and testing datasets.

4.6. LR model predictions

A model based on LR has also been developed in this investigation to predict the concrete's CS. The model was developed to create an equation based on the least square method (Eq. (17)).

$$\mathbf{CS}\ (\mathbf{MPa}) = -29.54 * \boldsymbol{w}/\boldsymbol{c} + 45.284 * \boldsymbol{w}/\boldsymbol{b} - 3.84 * \boldsymbol{A}/\boldsymbol{c} - 0.056 * \mathbf{RCA}(\%) + 100.004 \tag{17}$$

To assess the performance of the LR-based models, three statistical measures were used, namely CC, RMSE, and MAE. According to the results obtained, the LR-based models performed poorly, with CC = 0.691, 0.585, MAE = 6.058, 9.309, RMSE = 8.244, 11.467, and MAPE = 13.1%, 18.2% for the training and testing datasets, respectively (Refer to Table 5). The agreement plot in Fig. 12 shows that the predicted values of the LR-based model do not match the observed values very well. In other words, it is clear that the LR-based models performed poorly in this specific scenario due to the existence of noise and outliers in the data, which can damage the performance of the model.

4.7. ANNs model predictions

The ANN model was built by repetitive rounds of trial and error, such as "the number of hidden layer neurons, number of hidden layers, momentum, and learning rate". There are three hidden layers in the created ANN model. The hidden layers have ten neurons in

Table 3
VIF and the tolerance values between each one of the independent input variables and all the other independent variables simultaneously.

		Inputs							Outputs
		w/c	w/b	A/c	FA/CA	SP (kg)	w/s	RCA (%)	CS (MPa)
Collinearity Statistics	VIF	3.17	4.48	4.97	2.75	1.31	4.99	1.01	-
	1/VIF	0.17	0.12	0.11	0.20	0.51	0.11	0.91	-

1/VIF: the tolerance.

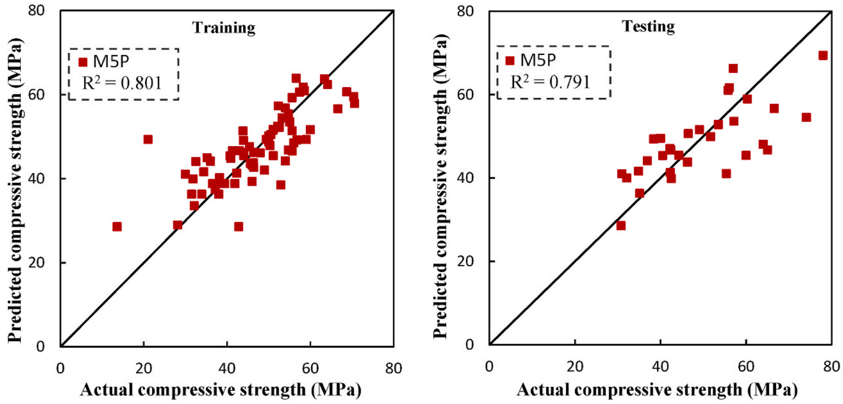

Fig. 9. Comparing observed and predicted CS of HP-SCC with RCA using an M5P-based model on training and testing datasets.

Table 4
The linear equations obtained through the utilization of the M5P-based model.

LM number	Linear Equations
1	$\boldsymbol{CS}\ (\boldsymbol{MPa}) = -71.771 * \boldsymbol{w/c} + 15.7967 * \boldsymbol{w/b} - 1.4842 * \boldsymbol{w/s} - 3.6368 * \boldsymbol{A/c} - 0.0926 * \boldsymbol{RCA}\ (\%) + 109.8769$
2	$\mathbf{CS}\ (\mathbf{MPa}) = -6.8165 * \mathbf{w/c} + 10.4501 * \mathbf{w/b} - 0.9818 * \mathbf{w/s} - 0.0501 - 0.8859 * \mathbf{A/c} - 4.5438 * \mathbf{FA/CA} + 1.9441 * \mathbf{SP}\ (\mathbf{kg}) + * \mathbf{RCA}\ (\%) + 61.7007$

Table 5
Performance evaluation indices of the models.

Models	Training Stage				Testing Stage				COM	Ranking
	CC	MAE	RMSE	MAPE	CC	MAE	RMSE	MAPE		
M5P-based Model	0.801	4.9	6.86	0.107	0.791	7.099	8.799	0.138	0.84	4
RF-based Model	0.966	2.212	3.23	0.048	0.887	5.044	6.951	0.098	0.45	2
SVM_PUK-based Model	0.894	1.721	5.137	0.045	0.9	3.813	6.306	0.076	0.36	1
LR-based Model	0.691	6.058	8.244	0.131	0.585	9.309	11.467	0.182	1.06	5
ANN-based Model	0.995	1.677	1.94	0.039	0.825	6.803	8.7	0.14	0.69	3

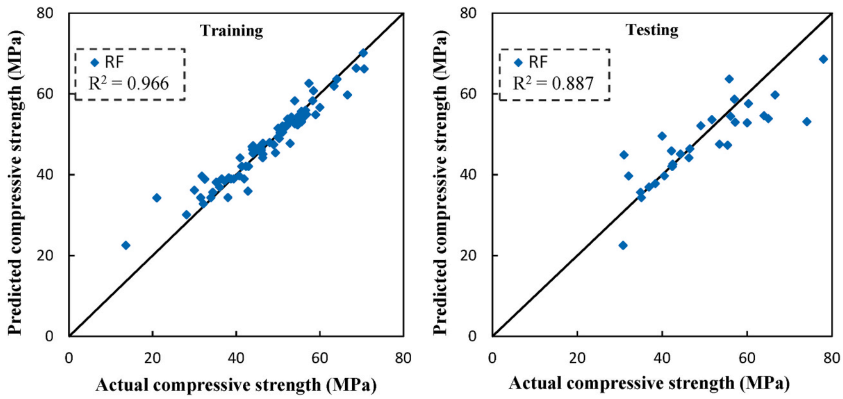

Fig. 10. Comparing observed and predicted CS of HP-SCC with RCA using an RF-based model on training and testing datasets.

the first and two neurons in both the second and third as illustrated in Fig. 13 with a momentum of 0.1 and a learning rate of 0.2. Table 6 shows the performance of the ANN-based model, which achieved a CC = 0.995, 0.825, MAE = 1.677, 6.803, RMSE = 1.940, 8.700, and MAPE = 3.9%, 14% (Refer to Table 5) for both stages (Training and Testing, respectively). Fig. 14, display agreement plots between observed and predicted CS in both stages (training and testing). The ANN-based model outperformed other models in the training stage owing to MAPE with a value of 3.9%. However, the performance was the second worst after the LR-based model with a MAPE of 14%.

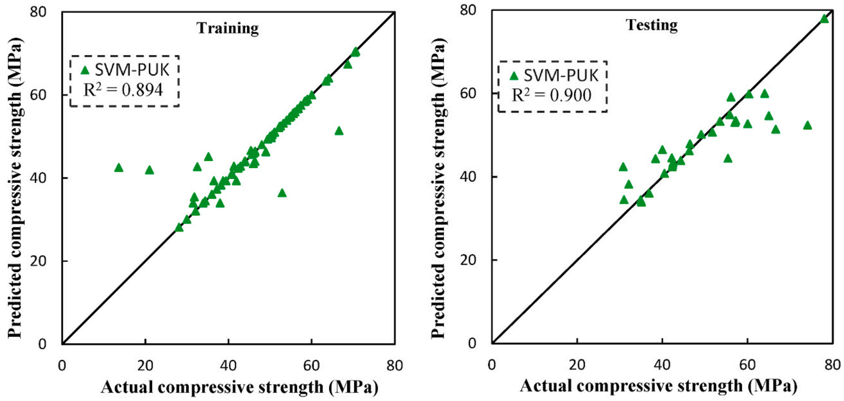

Fig. 11. Comparing observed and predicted CS of HP-SCC with RCA using SVM-PUK-based model on training and testing datasets.

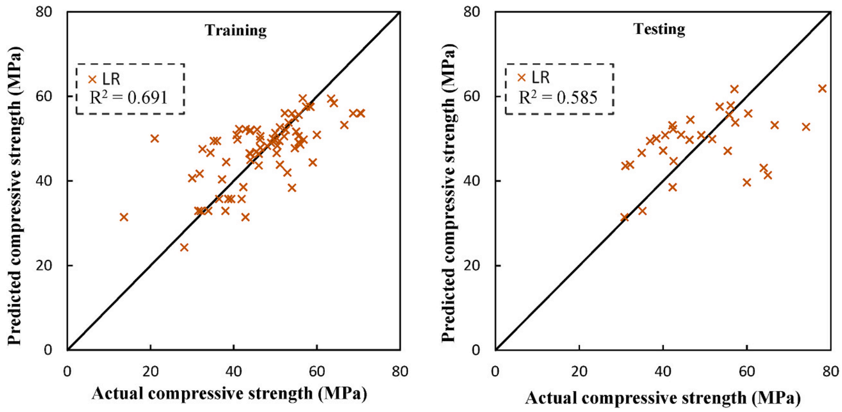

Fig. 12. Comparing observed and predicted CS of HP-SCC with RCA using the LR-based model on training and testing datasets.

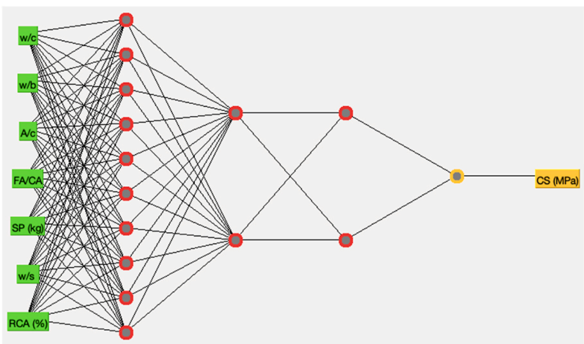

Fig. 13. ANN structures of the best model for CS.

4.8. Comparison of soft computing-based models to ANNs-based model

Fig. 15–17 present a comparative analysis of soft computing-based models, revealing that the SVM-PUK-based model exhibits superior performance compared to the ANNs-based model and other applied soft computing-based models. To evaluate the efficacy of multiple regression and soft computing-based models in predicting the CS of HP-SCC, errors and performances were assessed in Fig. 15 and Fig. 16 for both the training and testing phases. The results demonstrate that the SVM_PUK model outperforms other techniques, displayed in the evaluation of statistical indicators for the SVM_PUK-based Model, CC (Fig. 15a), MAE (Fig. 15b), RMSE (Fig. 15c), and SI (Fig. 16) for the testing datasets were equal to 0.900, 3.813, 6.306 MPa, 0.13 and respectively. However, the RF-based Model performed best in the training stage amongst other techniques, as shown in Fig. 16. SVM_PUK model is still the best predictive model, whereby the order was M5P_pruned, RF, LR, and ANN, respectively. It is incidental from the Radar plots that the SVM-PUK-based

Table 6
Comparison of the models developed in the current study with those existing in the literature.

References	Models to predict the CS (MPa)	Input No.	Input type	MAE	RSME	CC
[79]	$\overbrace{CS}^{*}_{cu} = -0.32 + 0.022 * \boldsymbol{RCA} + (1 - 0.025 * \boldsymbol{RCA}) * \boldsymbol{CS_{NCA}}$	7	Dimensional	-	-	0.70
[80]	$\overbrace{CS}^{*}_{cu} = \dfrac{19.1 \times 0.998^{\boldsymbol{RCA}} \times \left(\frac{\boldsymbol{w}}{\boldsymbol{c}} + 0.33\right)}{\left(\frac{\boldsymbol{w}}{\boldsymbol{c}}\right)^{1.5}}$	5	Dimensional	-	7.90	<0.40
[81]	$\overbrace{CS}^{*}_{cu} = \dfrac{28.97 - 4.71(RCA)^{1.69}}{\left(\frac{w}{c}\right)^{0.63}}$	4	Dimensional	-	11.75	<0.40
[82]	$\overbrace{CS}^{*}_{cu} = -0.0902 * \boldsymbol{W} + 0.1089 * \mathbf{A} + 0.1279 * \boldsymbol{C} + 0.1065 * \boldsymbol{F} + 0.3079 * \mathbf{S} + 0.0229 * \mathbf{CA} + 0.03116 * \mathbf{FA} - 50.77$	8	Dimensional	7.61	-	0.65
[83]	ANN	8	Dimensional	3.41	4.85	0.90
[84]	SVM	8	Dimensional	7.43	7.73	0.78
[85]	M5P	7	Dimensional	0.84	1.20	0.89
[83]	ANN	14	Dimensional and Non-dimensional	11.02	-	0.87
[83]	Tree-based	14	Dimensional and Non-dimensional	-	8.371	0.69
Current study	$\overbrace{CS}^{*}_{cu} = -29.54 * \boldsymbol{w/c} + 45.284 * \boldsymbol{w/b} - 3.84 * \boldsymbol{A/c} - 0.056 * \mathbf{RCA}(\%) + 100.004$	7	Non-dimensional	7.03	9.21	0.66
Current study	M5P with two LMs: 1. $\overbrace{CS}^{*}_{cu} = -71.771 * \boldsymbol{w/c} + 15.7967 * \boldsymbol{w/b} - 1.4842 * \boldsymbol{w/s} - 3.6368 * \boldsymbol{A/c} - 0.0926 * \boldsymbol{RCA}\ (\%) + 109.8769$ 2. $\overbrace{CS}^{*}_{cu} = -6.8165 * \mathbf{w/c} + 10.4501 * \mathbf{w/b} - 0.9818 * \mathbf{w/s} - 0.0501 - 0.8859 * \mathbf{A/c} - 4.5438 * \mathbf{FA/CA} + 1.9441 * \mathbf{SP}\ (\mathbf{kg}) + * \mathbf{RCA}\ (\%) + 61.7007$	7	Non-dimensional	5.56	7.44	0.80
Current study	RF	7	Non-dimensional	3.06	4.35	0.94
Current study	SVM	7	Non-dimensional	2.35	5.49	0.90
Current study	ANN	7	Non-dimensional	3.21	3.97	0.94

All Equations for volumetric replacement of 0–100% RCA, * for cubic specimens, CA/c: coarse aggregate to cement ratio, w/c: water to cement ratio, WA: water absorption of coarse aggregates, RCA: replacement ratio, C: cement content, W: water content, FA: fine aggregate, A: age of specimens, B: blast furnace slag content, F: fly ash content, S: superplasticizer content, CA: coarse aggregate content.

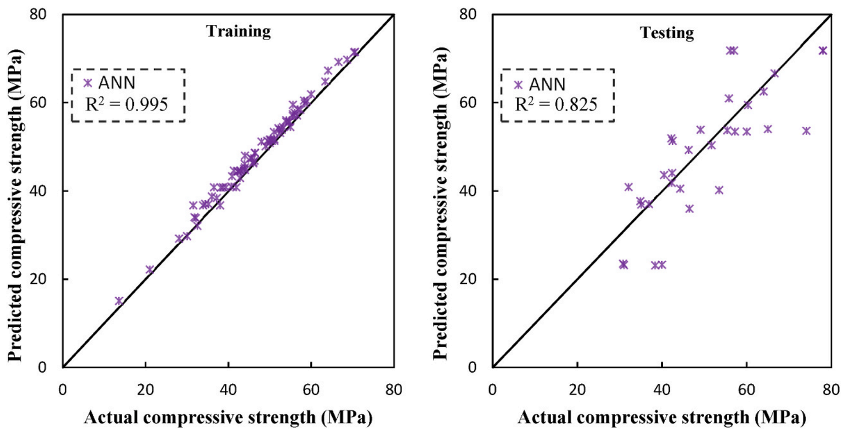

Fig. 14. Comparing observed and predicted CS of HP-SCC with RCA using the ANN-based model on training and testing datasets.

model's predicted values were close to the actual CS of HP-SCC, which endorses the findings of Huang et al. and Alyaseen et al. [34,70], whereas, the LR-based model was the worst. Even though the RF-based and ANN-based models performed better in the training stage, the SVM-PUK-based model outperformed them in the testing stage with higher accuracy and lower errors. Moreover, The aforementioned evaluations are consistent with the rankings of the models based on COM, as depicted in Table 5. In accordance with these results, the SVM-PUK-based model emerged as the top-performing approach, while the RF-based model exhibited a comparatively

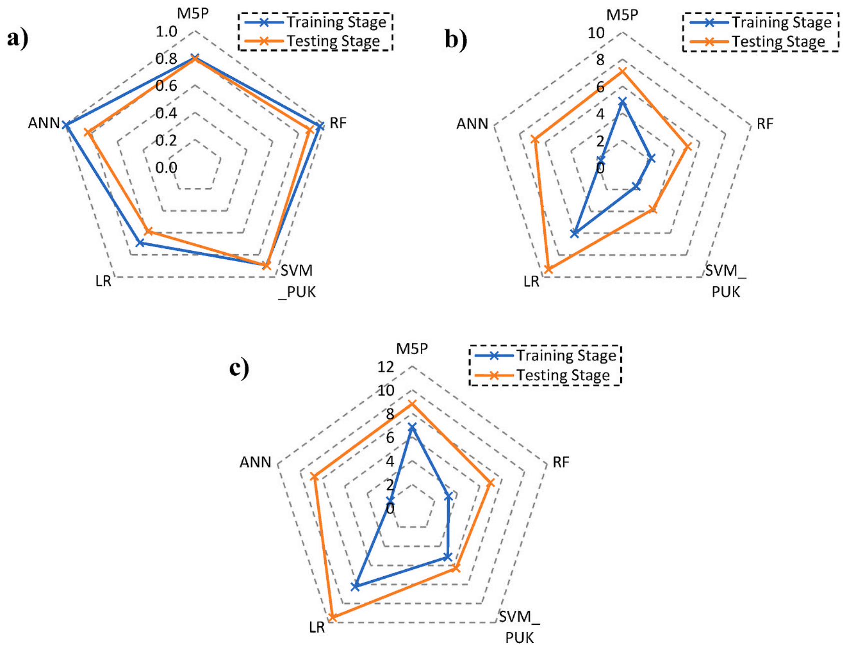

Fig. 15. Radar plot for model comparison based on: a) CC, b) MAE, c) RMSE.

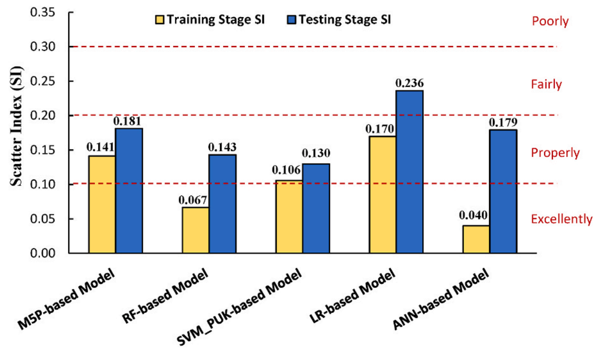

Fig. 16. Scatter Index plot for model comparison based on training and testing stages.

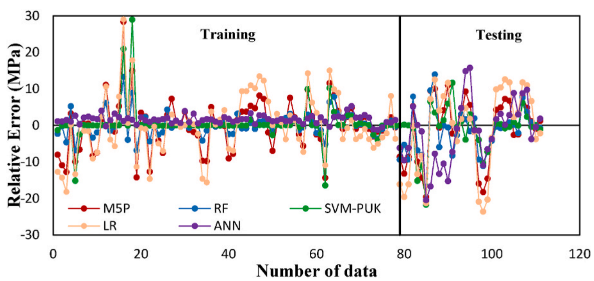

Fig. 17. CS relative error of all models using training and testing datasets.

weaker performance. Notably, the LR-based model demonstrated the poorest performance among the evaluated techniques. Finally, the differences in the relative errors of all techniques are shown in Fig. 17.

4.9. Comparing results with contemporary works

Furthermore, to assess the relative performance of the developed models against existing ones in the literature, we computed three statistical indicators - CC, MAE, and RMSE - utilizing the actual and predicted dataset (refer to Table 6). Upon comparing the model values from this study with those found in the literature, it is evident that all presented models exhibit higher CC values than their counterparts in prior research. Moreover, the LR and M5P linear models developed in this study outperform the models from the literature, demonstrating superior predictive capabilities for CS with CC values of 0.66 and 0.80, respectively.

5. Uncertainty analysis

Monte-Carlo simulation (MCS) is a valuable technique for assessing the uncertainty inherent in a model. By employing MCS, the authors can effectively capture the random nature of uncertainties associated with a model. In the context of predicting CS, it is crucial to acknowledge and address the existing uncertainties in prediction models, including uncertainties related to experimental, input predictors, and model parameters. To quantify these uncertainties, equations (18–20) are utilized to determine the individual error in CS prediction across the entire dataset. These equations calculate the mean ($\overline{E}$) and standard deviation (SD_E) of the computed error, providing insights into the magnitude and variability of the prediction errors [86].

$$E_i = log_{10}(CS_{pr}) - log_{10}(CS_{ac}) \tag{18}$$

$$\overline{E} = \sum_{i=1}^{n} E_i \tag{19}$$

$$SD_E = \sqrt{\sum_{i=1}^{n} \frac{(E_i - \overline{E})}{1 = n}} \tag{20}$$

Where: “n” represents the dataset’s length, while “CS_{pr}” and “CS_{ac}” represent the predicted and actual CS values, respectively.

In order to account for the elimination of the continuity correction process in the obtained errors, it becomes necessary to define a confidence range for the resultant errors. This requirement can be fulfilled by employing the Wilson score method, which utilizes the calculated values of " $\overline{E}$" and " SD_E". This method ensures the establishment of a confidence interval for the errors, providing a measure of the uncertainty associated with the predictions [87]. Furthermore, the Mean Absolute Deviation (MAD) can be determined using Eq. (21), enabling an assessment of the average deviation between the predicted values and the actual values. MAD is used to check the uncertainty of the model using Eq. (22) [87].

$$MAD = \frac{1}{n} \sum_{i=1}^{n} \left| CS_{pr} - \overline{CS}_{pr} \right| \tag{21}$$

$$Uncertainty = \frac{MAD * 100}{\overline{CS}_{pr}} \tag{22}$$

In the realm of uncertainty analysis, per [86], the presence of a negative mean value signifies an underestimation of the parameter during the prediction process. This implies that the predicted values fall short of the actual values, indicating a tendency toward underestimation. Conversely, a positive mean value indicates the degree of overestimation, suggesting that the predicted values surpass the actual values.

The uncertainty analysis results of the developed models in this study are presented in Table 7. MAD values for each model are expressed as a percentage of their respective median values. Specifically, the calculated percentages for the M5P, RF, SVM, LR, and ANN-based models are 21%, 20.7%, 19%, 22%, and 19%, respectively. According to Verbeeck et al. [88], these percentages fall within an acceptable range for estimating the uncertainty of models, as they are below the threshold of 35%.

6. Sensitivity analysis

In this study, the Cosine Amplitude Method (CAM) is employed to assess the impact of each variable on the predicted output. The sensitivity degree index associated with each input variable, as proposed by Yang and Zhang [89], is used to quantify this effect. By establishing a relationship between the input and output data, the sensitivity degree can be calculated using Eq. (23). To implement this technique, all data pairs were used to construct a data array denoted as $\boldsymbol{Y}$ following the procedure outlined below:

The input variable y_i in the array Y_i is a length vector of n as: $y_i = \{y_{i1}, y_{i2}, \ldots, y_{in}\}$ and the same for an output variable y_j in the array Y_j is a length vector of n as: $y_j = \{y_{j1}, y_{j2}, \ldots, y_{jn}\}$. Therefore, Eq. (23) shows the strength of CC_{ij} between the dataset Y_i and Y_j:

Table 7
Monte-Carlo-based uncertainty analysis of used models.

Model	$\overline{E}$	SD_E	Median	MAD	Uncertainty (%)
M5P	0.383	0.619	46.649	9.796	21.0
RF	0.173	0.416	47.310	9.770	20.7
SVM	0.592	0.769	46.475	8.830	19.0
LR	0.566	0.752	49.746	10.940	22.0
ANN	0.454	0.673	50.137	9.892	19.7

$$CC_{ij} = \frac{\sum_{k=1}^{n} y_{ik} \times y_{jk}}{\sqrt{\sum_{k=1}^{n} y_{ik}^{2} \times \sum_{k=1}^{n} y_{jk}^{2}}}, \textit{whenever } CC_{ij} \textit{is close to } 1 \therefore \textit{strong correlation} \tag{23}$$

Based on the results obtained from comparing the actual and predicted values, Fig. 18 illustrates the level of importance for the variables and their respective impact on the CS of HP-SCC. It is evident that the CC_{ij} values in all models closely align with the experimental values. Notably, the results indicate that the variables w/c, w/b, and w/s exert the most significant influence on the CS of HP-SCC, while the RCA(%) variable has the least impact. According to Alyaseen et al., w/c, w/b, and w/s were the most critical inputs in the CS prediction, which endorses the findings of this work.

7. Limitations of the work

Previous studies employed different testing methods, irrespective of the sample type, texture, and size of the aggregates. Additionally, the shape of the aggregates used in previous works consisted of both sharp and rounded edges. Furthermore, the size of the aggregates varied between 10 and 30 mm. Conversely, this research focused explicitly on cubic samples with a size of 150 mm, utilizing aggregates of up to 20 mm in size. One notable concern regarding this study is the limited availability of data, which can impact the accuracy and utility of machine learning algorithms. A smaller dataset increases the likelihood of biased or excessively favorable results, which can undermine the credibility of the study's conclusions [90,91]. However, even with a large dataset, accurately predicting outcomes using machine learning models can be challenging due to the complex relationships between input and output variables [92]. In the present work, the authors addressed these potential issues by selecting parameters that align with previous studies that employed similar materials, sample types, aggregate shapes, and sizes. Furthermore, ensured the absence of collinearity among independent variables in the selected dataset (For only: up to 100% utilization of RCA, the existence of SCMs along with chemical admixtures, 150 mm cubic samples, up to 20 mm aggregate size, and sharp edges aggregates).

8. Conclusion

This research employed a systematic data analytic methodology to analyze and evaluate experimental datasets obtained from HP-SCC samples with different RCA and SCMs ratios. The primary objective was to assess the CS of concrete by examining the analysis indicators. A systematic framework was proposed to achieve this goal, and multiscale methods were utilized to obtain comprehensive and in-depth insights into the dataset, including PCC, M5P, RF, SVM, LR, and ANNs. Based on an analysis and modeling of datasets, it is conceivable to extract the following conclusions regarding the prediction of the CS of HP-SCC mixes:

- The proportion of RCA substitute with NCA (fine or coarse aggregate) ranged from 0 to 100%. Wherein, the average ratio of RCA employed in producing HP-SCC mixes was 40%. The curing time for data from the literature was 28 days.
- The evaluation started with a correlation analysis, in which PCC analysis was implemented. Within the same variable category, variable groups were identified in which the correlation between each pair of variables was robust. It was clear that all w/b & w/s, FA/CA & A/c, w/c & w/s, and w/b & w/c respectively >0.5, which reveals that whenever one variable changes, the other variable changes in the exact direction strongly.
- In addition to the positive correlations, some variables (w/s & A/c, w/b & A/c, w/c & CS, and FA/CA & w/s) had relatively strong negative correlations between some pairs of variables > -0.5. However, the correlations between the rest of the variables (w/c & A/c, FA/CA & w/c, SP & w/c, RCA & w/c, RCA & SP, RCA & A/c, RCA & w/s, CS & A/c) were just normal < -0.5 (there is no relationship or slightly related).

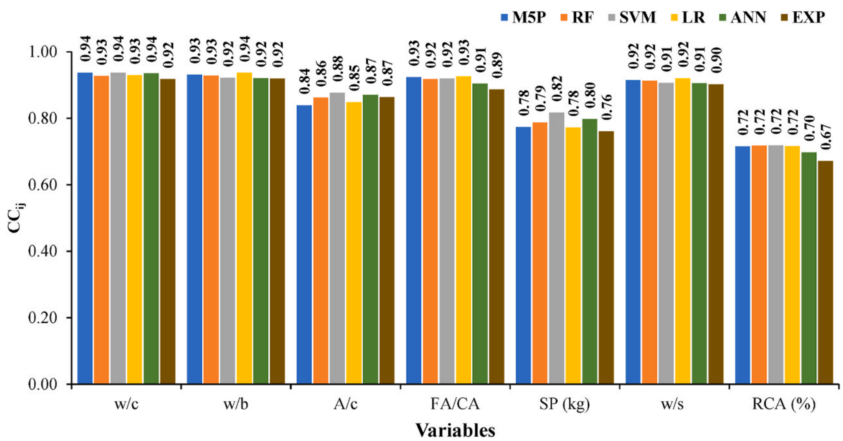

Fig. 18. Results of the conducted sensitivity analysis through CC_{ij} factor.

- Based on VIF, there is no evidence of multicollinearity among the independent input variables in the dataset. This means that the independent input variables are not highly correlated with each other. However, being VIF close to 5 might not affect the accuracy of predictive models and potentially introduce bias but explains the importance of the inputs in the sensitivity analysis. Thus, these findings suggest that the variables with high correlation should be carefully considered, which might be the reason behind being w/c and w/b are the most important variables of the sensitivity analysis in the literature. The results of VIF and tolerance were as w/c = 3.17, 0.17, w/b = 4.48, 0.12, A/c = 4.97, 0.11, FA/CA = 2.75, 0.2, SP = 1.31, 0.51, w/s = 4.99, 0.11, RCA = 1.01, 0.91, respectively.
- The present study employed the M5P, RF, SVM, LR, and ANN models to predict the CS of HP-SCC. The performance of these models was evaluated using various assessment criteria. The results indicate that the SVM-PUK model exhibited superior predictive ability over the other models, as evidenced by its higher CC, minor RMS, MAE, and MAPE values (0.900, 3.813, 6.306, 0.076 respectively). Conversely, based on the data examined in this study, the LR model performed poorly in its ability to predict the CS of HP-SCC. Thus, the SVM-PUK model is the most appropriate and effective tool for predicting the CS of HP-SCC in this context.
- SVM, RF, ANN, M5P, and LR are the suggested models in descending order of practicality, accuracy, and supremacy to predict the CS of HP-SCC with accordance to COM of 0.36, 0.45, 0.69, 0.84, 1.06, respectively.
- Based on the SI, it is clear that however, the RF-based model was best for predicting CS in the training stage and excellently performed (SI = 0.067), SVM-PUK-based model has a balanced performance in the training and testing stages and properly predicted the CS (SI = 0.106, 0.130).
- MAD for each specific model is expressed as a percentage relative to its median value. Specifically, the M5P, RF, SVM, LR, and ANN-based models exhibit MAD percentages of 21%, 20.7%, 19%, 22%, and 19%, respectively. Notably, all uncertainty values below 35% fall within the acceptable range in conducted analysis.
- The sensitivity analysis conducted in this study revealed that the variables in order from w/c, w/b, and w/s exhibit the most influencing parameters on CS of HP-SCC, while the RCA(%) variable has the least impact.

The overall conclusions revealed that vast quantities of RCA and industrial by-products can be effectively utilized in the production of sustainable HP-SCC, resulting in concrete with acceptable CS. Whereby, the optimal RCA content highly depends on RCA's shape, type, and surface texture not only the ratio of substitution. Thus, the aforementioned parameters are recommended to be added to the future research for the best understanding of their effects.

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CRediT authorship contribution statement

Ahmad Alyaseen: Conceptualization, Methodology, Formal analysis, Resources, Writing – review & editing, Writing – original draft, Project administration, Visualization, and, Writing – review & editing. **Arunava Poddar:** Supervision, Writing – original draft, Visualization, Writing – review & editing. **Navsal Kumar:** Formal analysis, Resources, Writing – review & editing. **Salwan Tajjour:** Python coding. **C. Venkata Siva Rama Prasad:** Methodology, Formal analysis, Resources, Writing – review & editing. **Hussain Alahmad:** Conceptualization, Writing – review & editing. **Parveen Sihag:** Supervision, Writing – review & editing, read and approved the final manuscript. The authors confirm that all individuals involved in the research have thoroughly read and given their consent to the final version of the manuscript that is to be published.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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References

[1] P.T. Anastas, J.B. Zimmerman, Chapter 2 the twelve principles of green engineering as a foundation for sustainability, Sustain. Sci.Eng 1 (C) (Jan. 2006) 11–32, https://doi.org/10.1016/S1871-2711(06)80009-7.

[2] C. Meyer, The greening of the concrete industry, Cem. Concr. Compos. 31 (8) (Sep. 2009) 601–605, https://doi.org/10.1016/j.cemconcomp.2008.12.010.

[3] A.K. Padmini, K. Ramamurthy, M.S. Mathews, Influence of parent concrete on the properties of recycled aggregate concrete, Construct. Build. Mater. 23 (2) (Feb. 2009) 829–836, https://doi.org/10.1016/j.conbuildmat.2008.03.006.

[4] A. Abbas, et al., Quantification of the residual mortar content in recycled concrete aggregates by image analysis, Mater. Char. 60 (7) (Jul. 2009) 716–728, https://doi.org/10.1016/J.MATCHAR.2009.01.010.

[5] R.V. Silva, J. de Brito, R.K. Dhir, Availability and processing of recycled aggregates within the construction and demolition supply chain: a review, J. Clean. Prod. 143 (Feb. 2017) 598–614, https://doi.org/10.1016/J.JCLEPRO.2016.12.070.

[6] A. Gonzalez-Corominas, M. Etxeberria, I. Fernandez, Structural behaviour of prestressed concrete sleepers produced with high performance recycled aggregate concrete, Materials and Structures/Materiaux et Constructions 50 (1) (Feb. 2017) 1–14, https://doi.org/10.1617/S11527-016-0966-6/FIGURES/10.

[7] C. Shi, Z. Wu, K. Lv, L. Wu, A review on mixture design methods for self-compacting concrete, Construct. Build. Mater. 84 (Jun. 2015) 387–398, https://doi.org/10.1016/J.CONBUILDMAT.2015.03.079.

[8] N. Makul, Modern sustainable cement and concrete composites: review of current status, challenges and guidelines, Sustain.Mater. Technol. 25 (Sep. 2020), e00155, https://doi.org/10.1016/J.SUSMAT.2020.E00155.

[9] L. Evangelista, J. de Brito, Mechanical behaviour of concrete made with fine recycled concrete aggregates, Cem. Concr. Compos. 29 (5) (May 2007) 397–401, https://doi.org/10.1016/j.cemconcomp.2006.12.004.

[10] A. Rao, K.N. Jha, S. Misra, Use of aggregates from recycled construction and demolition waste in concrete, Resour. Conserv. Recycl. 50 (1) (Mar. 2007) 71–81, https://doi.org/10.1016/j.resconrec.2006.05.010.

[11] S. Arora, B. Singh, B. Bhardwaj, Strength performance of recycled aggregate concretes containing mineral admixtures and their performance prediction through various modeling techniques, J. Build. Eng. 24 (Jul. 2019), 100741, https://doi.org/10.1016/J.JOBE.2019.100741.

[12] J. Sobhani, M. Khanzadi, A.H. Movahedian, Support vector machine for prediction of the compressive strength of no-slump concrete, Comput. Concr. 11 (4) (2013) 337–350, https://doi.org/10.12989/CAC.2013.11.4.337.

[13] B.A. Omran, Q. Chen, R. Jin, Comparison of data mining techniques for predicting compressive strength of environmentally friendly concrete, J. Comput. Civ. Eng. 30 (6) (Nov. 2016), 04016029, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000596.

[14] D.C. Feng, et al., Machine learning-based compressive strength prediction for concrete: an adaptive boosting approach, Construct. Build. Mater. 230 (Jan. 2020), 117000, https://doi.org/10.1016/J.CONBUILDMAT.2019.117000.

[15] R.H. Faraj, H.U. Ahmed, S. Rafiq, N.H. Sor, D.F. Ibrahim, S.M.A. Qaidi, Performance of Self-Compacting mortars modified with Nanoparticles: a systematic review and modeling, Cleaner Materials 4 (Jun. 2022), 100086, https://doi.org/10.1016/J.CLEMA.2022.100086.

[16] J. Sobhani, M. Najimi, A.R. Pourkhorshidi, T. Parhizkar, Prediction of the compressive strength of no-slump concrete: a comparative study of regression, neural network and ANFIS models, Construct. Build. Mater. 24 (5) (May 2010) 709–718, https://doi.org/10.1016/J.CONBUILDMAT.2009.10.037.

[17] Y. Yu, W. Li, J. Li, T.N. Nguyen, A novel optimised self-learning method for compressive strength prediction of high performance concrete, Construct. Build. Mater. 184 (Sep. 2018) 229–247, https://doi.org/10.1016/j.conbuildmat.2018.06.219.

[18] Q.B. Pham, et al., Groundwater level prediction using machine learning algorithms in a drought-prone area, Neural Comput. Appl. 34 (13) (Jul. 2022) 10751–10773, https://doi.org/10.1007/S00521-022-07009-7/TABLES/7.

[19] I.C. Yeh, L.C. Lien, Knowledge discovery of concrete material using Genetic Operation Trees, Expert Syst. Appl. 36 (3) (Apr. 2009) 5807–5812, https://doi.org/10.1016/J.ESWA.2008.07.004.

[20] U. Atici, Prediction of the strength of mineral admixture concrete using multivariable regression analysis and an artificial neural network, Expert Syst. Appl. 38 (8) (Aug. 2011) 9609–9618, https://doi.org/10.1016/J.ESWA.2011.01.156.

[21] S. Paruthi, A. Husain, P. Alam, A. Husain Khan, M. Abul Hasan, H.M. Magbool, A review on material mix proportion and strength influence parameters of geopolymer concrete: application of ANN model for GPC strength prediction, Construct. Build. Mater. 356 (Nov. 2022), 129253, https://doi.org/10.1016/j.conbuildmat.2022.129253.

[22] R. Solhmirzaei, H. Salehi, V. Kodur, M.Z. Naser, Machine learning framework for predicting failure mode and shear capacity of ultra high performance concrete beams, Eng. Struct. 224 (Dec. 2020), 111221, https://doi.org/10.1016/J.ENGSTRUCT.2020.111221.

[23] E. Garcia-Taengua, M. Bakhshi, L. Ferrara, Meta-analysis of steel fiber-reinforced concrete mixtures leads to practical mix design methodology, Materials 14 (14) (Jul. 2021) 3900, https://doi.org/10.3390/MA14143900/S1.

[24] M.M. Kamruzzaman, et al., Wireless Communication Channel Modeling Using Machine Learning, Transactions on Emerging Telecommunications Technologies, 2022, p. e4661, https://doi.org/10.1002/ETT.4661.

[25] J.C. Egbueri, J.C. Agbasi, Data-driven soft computing modeling of groundwater quality parameters in southeast Nigeria: comparing the performances of different algorithms, Environ. Sci. Pollut. Control Ser. 29 (25) (Jan. 2022) 38346–38373, https://doi.org/10.1007/S11356-022-18520-8, 2022 29:25.

[26] M. Adaikkappan, N. Sathiyamoorthy, Modeling, state of charge estimation, and charging of lithium-ion battery in electric vehicle: a review, Int. J. Energy Res. 46 (3) (Mar. 2022) 2141–2165, https://doi.org/10.1002/ER.7339.

[27] I. Ramezani, K. Moshkbar-Bakhshayesh, N. Vosoughi, M.B. Ghofrani, Applications of Soft Computing in nuclear power plants: a review, Prog. Nucl. Energy 149 (Jul. 2022), 104253, https://doi.org/10.1016/J.PNUCENE.2022.104253.

[28] E. Soranzo, C. Guardiani, W. Wu, A soft computing approach to tunnel face stability in a probabilistic framework, Acta Geotech 17 (4) (Apr. 2022) 1219–1238, https://doi.org/10.1007/S11440-021-01240-7/TABLES/7.

[29] T.-D. Nguyen, R. Cherif, P.-Y. Mahieux, J. Lux, A. Aït-Mokhtar, E. Bastidas-Arteaga, Artificial intelligence algorithms for prediction and sensitivity analysis of mechanical properties of recycled aggregate concrete: a review, J. Build. Eng. 66 (May 2023), 105929, https://doi.org/10.1016/j.jobe.2023.105929.

[30] H. Dabiri, M. Kioumarsi, A. Kheyroddin, A. Kandiri, F. Sartipi, Compressive strength of concrete with recycled aggregate; a machine learning-based evaluation, Cleaner Materials 3 (Mar. 2022), 100044, https://doi.org/10.1016/j.clema.2022.100044.

[31] I. Nunez, A. Marani, M. Flah, M.L. Nehdi, Estimating compressive strength of modern concrete mixtures using computational intelligence: a systematic review, Construct. Build. Mater. 310 (Dec. 2021), 125279, https://doi.org/10.1016/j.conbuildmat.2021.125279.

[32] W. Ben Chaabene, M. Flah, M.L. Nehdi, Machine learning prediction of mechanical properties of concrete: critical review, Construct. Build. Mater. 260 (Nov. 2020), 119889, https://doi.org/10.1016/j.conbuildmat.2020.119889.

[33] S. Devi, F. Esmaeilbeiki, · Sahar, M. Karimi, P. Sihag, Prediction of sulphur dioxide (SO2) in air by using bagging, ANN and M5P: a case study, Gaya and Tirupati, India, Arabian J. Geosci. 15 (7) (Mar. 2022) 1–16, https://doi.org/10.1007/S12517-022-09725-9, 2022 15:7.

[34] A. Alyaseen, et al., Application of soft computing techniques for the prediction of splitting tensile strength in bacterial concrete, J.Struct. Integr.Mainten 8 (1) (Jan. 2023) 26–35, https://doi.org/10.1080/24705314.2022.2142900.

[35] F.H. Almohammed, A. Upadhya, A. Alyaseen, Using soft computing techniques to predict the values of compressive strength of concrete with basalt fiber reinforced concrete, Appl.Comput. Intell. Concr. Technol. (Jun. 2022) 183–206, https://doi.org/10.1201/9781003184331-11.

[36] F.H. Almohammed, A. Alyaseen, A. Upadhya, Using regression model to estimate the splitting tensile strength for the concrete with basalt fiber reinforced concrete, Appl.Comput. Intell. Concr. Technol. (Jun. 2022) 143–161, https://doi.org/10.1201/9781003184331-9.

[37] A. Sepahvand, B. Singh, P. Sihag, A. Nazari Samani, H. Ahmadi, S. Fiz Nia, Assessment of the Various Soft Computing Techniques to Predict Sodium Absorption Ratio (SAR), 10.1080/09715010.2019.1595185, vol. 27, no. S1, pp. 124–135, 2019, https://doi.org/10.1080/09715010.2019.1595185.

[38] A. Mohammed, S. Rafiq, P. Sihag, W. Mahmood, K. Ghafor, W. Sarwar, ANN, M5P-tree model, and nonlinear regression approaches to predict the compression strength of cement-based mortar modified by quicklime at various water/cement ratios and curing times, Arabian J. Geosci. 13 (22) (Nov. 2020) 1–16, https://doi.org/10.1007/S12517-020-06199-5/TABLES/4.

[39] A. Mohammed, S. Rafiq, P. Sihag, R. Kurda, W. Mahmood, Soft computing techniques: systematic multiscale models to predict the compressive strength of HVFA concrete based on mix proportions and curing times, J. Build. Eng. 33 (Jan. 2021), 101851, https://doi.org/10.1016/J.JOBE.2020.101851.
[40] B. Singh, P. Sihag, K. Singh, Modelling of impact of water quality on infiltration rate of soil by random forest regression, Model Earth Syst Environ 3 (3) (Sep. 2017) 999–1004, https://doi.org/10.1007/S40808-017-0347-3/TABLES/5.
[41] A. Elbeltagi, et al., Modelling daily reference evapotranspiration based on stacking hybridization of ANN with meta-heuristic algorithms under diverse agro-climatic conditions, Stoch. Environ. Res. Risk Assess. 36 (10) (Oct. 2022) 3311–3334, https://doi.org/10.1007/S00477-022-02196-0/FIGURES/12.
[42] A.-D. Pham, N.-T. Ngo, T.T. Ha Truong, N.-T. Huynh, N.-S. Truong, Predicting energy consumption in multiple buildings using machine learning for improving energy efficiency and sustainability, J. Clean. Prod. 260 (Jul. 2020), 121082, https://doi.org/10.1016/j.jclepro.2020.121082.
[43] V.F. Rodriguez-Galiano, B. Ghimire, J. Rogan, M. Chica-Olmo, J.P. Rigol-Sanchez, An assessment of the effectiveness of a random forest classifier for land-cover classification, ISPRS J. Photogrammetry Remote Sens. 67 (Jan. 2012) 93–104, https://doi.org/10.1016/j.isprsjprs.2011.11.002.
[44] D.N. Mhawi, A. Aldallal, S. Hassan, Advanced feature-selection-based hybrid ensemble learning algorithms for network intrusion detection systems, Symmetry (Basel) 14 (7) (Jul. 2022) 1461, https://doi.org/10.3390/sym14071461.
[45] V.Y. Kulkarni, P.K. Sinha, Pruning of Random Forest classifiers: a survey and future directions, in: 2012 International Conference on Data Science & Engineering (ICDSE), IEEE, Jul. 2012, pp. 64–68, https://doi.org/10.1109/ICDSE.2012.6282329.
[46] Z. Wang, Y. Wang, R. Zeng, R.S. Srinivasan, S. Ahrentzen, Random Forest based hourly building energy prediction, Energy Build. 171 (Jul. 2018) 11–25, https://doi.org/10.1016/j.enbuild.2018.04.008.
[47] V. Vapnik, O. Chapelle, Bounds on error expectation for support vector machines, Neural Comput. 12 (9) (2000) 2013–2036, https://doi.org/10.1162/089976600300015042.
[48] O. Chapelle, P. Haffner, V.N. Vapnik, Support vector machines for histogram-based image classification, IEEE Trans. Neural Network. 10 (5) (1999) 1055–1064, https://doi.org/10.1109/72.788646.
[49] A. Afzal, S. Alshahrani, A. Alrobaian, A. Buradi, S.A. Khan, Power plant energy predictions based on thermal factors using ridge and support vector regressor algorithms, Energies 14 (21) (Nov. 2021) 7254, https://doi.org/10.3390/en14217254.
[50] S. Jha, E. Yang, A.O. Almagrabi, A.K. Bashir, G.P. Joshi, Retracted article: comparative analysis of time series model and machine testing systems for crime forecasting, Neural Comput. Appl. 33 (17) (Sep. 2021) 10621–10636, https://doi.org/10.1007/s00521-020-04998-1.
[51] Y. Bao, T. Xiong, Z. Hu, Multi-step-ahead time series prediction using multiple-output support vector regression, Neurocomputing 129 (Apr. 2014) 482–493, https://doi.org/10.1016/j.neucom.2013.09.010.
[52] A. Alyaseen, A. Poddar, F. Almohammed, S. Tajjour, K. Hammadeh, H. Alahmad, Compressive strength prediction and analysis of concrete using hybrid artificial neural networks, Appl.Comput. Intell. Concr. Technol. (Jun. 2022) 285–303, https://doi.org/10.1201/9781003184331-17.
[53] A. Alyaseen, A. Poddar, F. Almohammed, S. Tajjour, K. Hammadeh, H. Alahmad, Predicting recycled aggregates compressive strength in high-performance concrete using artificial neural networks, Appl.Comput. Intell. Concr. Technol. (Jun. 2022) 269–283, https://doi.org/10.1201/9781003184331-16.
[54] M.R.R. Kooh, R. Thotagamuge, Y.-F. Chou Chau, A.H. Mahadi, C.M. Lim, Machine learning approaches to predict adsorption capacity of Azolla pinnata in the removal of methylene blue, J. Taiwan Inst. Chem. Eng. 132 (Mar. 2022), 104134, https://doi.org/10.1016/j.jtice.2021.11.001.
[55] T. Khan, J. Westin, M. Dougherty, Classification of speech intelligibility in Parkinson's disease, Biocybern. Biomed. Eng. 34 (1) (2014) 35–45, https://doi.org/10.1016/j.bbe.2013.10.003.
[56] M.F.M. Zain, S.M. Abd, Multiple regression model for compressive strength prediction of high performance concrete, J. Appl. Sci. 9 (1) (Dec. 2008) 155–160, https://doi.org/10.3923/jas.2009.155.160.
[57] B. Kumar, N. Kumar, A. Elbeltagi, F.H. Almohammed, Evaluation of ANN and tree-based techniques for predicting the compressive strength of granite powder reinforced concrete, Appl.Comput. Intell. Concr. Technol. (Jun. 2022) 253–267, https://doi.org/10.1201/9781003184331-15.
[58] A. Elbeltagi, N. Kumar, A. Chandel, A. Arshad, C.B. Pande, A.R.M.T. Islam, Modelling the reference crop evapotranspiration in the Beas-Sutlej basin (India): an artificial neural network approach based on different combinations of meteorological data, Environ. Monit. Assess. 194 (3) (Mar. 2022) 1–20, https://doi.org/10.1007/S10661-022-09812-0/TABLES/10.
[59] I. Mokashi, et al., Nusselt number analysis from a battery pack cooled by different fluids and multiple back-propagation modelling using feed-forward networks, Int. J. Therm. Sci. 161 (Mar. 2021), 106738, https://doi.org/10.1016/j.ijthermalsci.2020.106738.
[60] A. Afzal, K.M. Yashawantha, N. Aslfattahi, R. Saidur, R.K. Abdul Razak, R. Subbiah, Back propagation modeling of shear stress and viscosity of aqueous Ionic-MXene nanofluids, J. Therm. Anal. Calorim. 145 (4) (Aug. 2021) 2129–2149, https://doi.org/10.1007/s10973-021-10743-0.
[61] A. Afzal, J.K. Bhutto, A. Alrobaian, A. Razak Kaladgi, S.A. Khan, Modelling and computational experiment to obtain optimized neural network for battery thermal management data, Energies 14 (21) (Nov. 2021) 7370, https://doi.org/10.3390/en14217370.
[62] A. Majdi, M. Beiki, Evolving neural network using a genetic algorithm for predicting the deformation modulus of rock masses, Int. J. Rock Mech. Min. Sci. 47 (2) (Feb. 2010) 246–253, https://doi.org/10.1016/J.IJRMMS.2009.09.011.
[63] H. Apaydin, H. Feizi, M.T. Sattari, M.S. Colak, S. Shamshirband, K.W. Chau, Comparative analysis of recurrent neural network architectures for reservoir inflow forecasting, Water 12 (5) (May 2020) 1500, https://doi.org/10.3390/W12051500, 2020, Vol. 12, Page 1500.
[64] W.H. Al Yamani, D.M. Ghunimat, M.M. Bisharah, Modeling and predicting the sensitivity of high-performance concrete compressive strength using machine learning methods, Asian J.Civ. Eng. (Mar. 2023), https://doi.org/10.1007/s42107-023-00614-4.
[65] J. Adler, I. Parmryd, Quantifying colocalization by correlation: the Pearson correlation coefficient is superior to the Mander's overlap coefficient, Cytometry Part A 77A (8) (Aug. 2010) 733–742, https://doi.org/10.1002/CYTO.A.20896.
[66] L. Ceballos-Laita, et al., Relationship between hip abductor muscle strength and frontal plane kinematics: a cross-sectional study in elite handball athletes, Appl. Sci. 12 (19) (Oct. 2022), 10044, https://doi.org/10.3390/APP121910044, 2022, Vol. 12, Page 10044.
[67] J. Fox, Applied Regression Analysis and Generalized Linear Models, Sage Publications., 2015 [Online]. Available: https://socialsciences.mcmaster.ca/jfox/Books/Applied-Regression-3E/chap-25.pdf. (Accessed 11 June 2023).
[68] W. Black, B.J. Babin, Multivariate data analysis: its approach, evolution, and impact, in: The Great Facilitator, Springer International Publishing, Cham, 2019, pp. 121–130, https://doi.org/10.1007/978-3-030-06031-2_16.
[69] J.H. Stapleton, Linear Statistical Model, second ed., John Wiley & Sons, 2009 [Online]. Available: https://www.wiley.com/en-us/Linear+Statistical+Models%2C+2nd+Edition-p-9780470231463. (Accessed 11 June 2023).
[70] X. Huang, M. Wasouf, J. Sresakoolchai, S. Kaewunruen, Prediction of healing performance of autogenous healing concrete using machine learning, Materials 14 (15) (Jul. 2021) 4068, https://doi.org/10.3390/ma14154068.
[71] K. Liu, J. Zheng, S. Dong, W. Xie, X. Zhang, Mixture optimization of mechanical, economical, and environmental objectives for sustainable recycled aggregate concrete based on machine learning and metaheuristic algorithms, J. Build. Eng. 63 (Jan. 2023), 105570, https://doi.org/10.1016/j.jobe.2022.105570.
[72] A. Alyaseen, A. Poddar, H. Alahmad, N. Kumar, P. Sihag, High-performance self-compacting concrete with recycled coarse aggregate: comprehensive systematic review on mix design parameters, J.Struct. Integr.Mainten (May 2023) 1–18, https://doi.org/10.1080/24705314.2023.2211850.
[73] R. Kurda, J. de Brito, J.D. Silvestre, Indirect evaluation of the compressive strength of recycled aggregate concrete with high fly ash ratios, Mag. Concr. Res. 70 (4) (Feb. 2018) 204–216, https://doi.org/10.1680/jmacr.17.00216.
[74] A. Mohammed, et al., ANN, M5P-tree and nonlinear regression approaches with statistical evaluations to predict the compressive strength of cement-based mortar modified with fly ash, J. Mater. Res. Technol. 9 (6) (Nov. 2020) 12416–12427, https://doi.org/10.1016/j.jmrt.2020.08.083.
[75] Y. Xu, et al., Computation of high-performance concrete compressive strength using standalone and ensembled machine learning techniques, Materials 14 (22) (Nov. 2021) 7034, https://doi.org/10.3390/ma14227034.
[76] L. Bu, G. Du, Q. Hou, Prediction of the compressive strength of recycled aggregate concrete based on artificial neural network, Materials 14 (14) (Jul. 2021) 3921, https://doi.org/10.3390/ma14143921.
[77] K. Khan, W. Ahmad, M.N. Amin, F. Aslam, A. Ahmad, M.A. Al-Faiad, Comparison of prediction models based on machine learning for the compressive strength estimation of recycled aggregate concrete, Materials 15 (10) (May 2022) 3430, https://doi.org/10.3390/ma15103430.

[78] J. de-Prado-Gil, C. Palencia, N. Silva-Monteiro, R. Martínez-García, To predict the compressive strength of self compacting concrete with recycled aggregates utilizing ensemble machine learning models, Case Stud. Constr. Mater. 16 (Jun. 2022), e01046, https://doi.org/10.1016/j.cscm.2022.e01046.
[79] C. Thomas, J. Setién, J.A. Polanco, P. Alaejos, M. Sánchez de Juan, Durability of recycled aggregate concrete, Construct. Build. Mater. 40 (Mar. 2013) 1054–1065, https://doi.org/10.1016/j.conbuildmat.2012.11.106.
[80] A. Gholampour, A.H. Gandomi, T. Ozbakkaloglu, New formulations for mechanical properties of recycled aggregate concrete using gene expression programming, Construct. Build. Mater. 130 (Jan. 2017) 122–145, https://doi.org/10.1016/j.conbuildmat.2016.10.114.
[81] J. Xu, X. Zhao, Y. Yu, T. Xie, G. Yang, J. Xue, Parametric sensitivity analysis and modelling of mechanical properties of normal- and high-strength recycled aggregate concrete using grey theory, multiple nonlinear regression and artificial neural networks, Construct. Build. Mater. 211 (Jun. 2019) 479–491, https://doi.org/10.1016/j.conbuildmat.2019.03.234.
[82] S.M. Mousavi, P. Aminian, A.H. Gandomi, A.H. Alavi, H. Bolandi, A new predictive model for compressive strength of HPC using gene expression programming, Adv. Eng. Software 45 (1) (Mar. 2012) 105–114, https://doi.org/10.1016/J.ADVENGSOFT.2011.09.014.
[83] N. Deshpande, S. Londhe, S. Kulkarni, Modeling compressive strength of recycled aggregate concrete by artificial neural network, model tree and non-linear regression, Int. J. Sustain. Built. Environ. 3 (2) (Dec. 2014) 187–198, https://doi.org/10.1016/j.ijsbe.2014.12.002.
[84] R. Mustapha, E.A. Mohamed, High-performance concrete compressive strength prediction based weighted support vector machines, J. Eng. Res.Appl 7 (1) (Jan. 2017) 68–75 [Online]. Available: https://www.ijera.com/papers/Vol7_issue1/Part-1/L0701016875.pdf. (Accessed 14 January 2023).
[85] A.R. Ghanizadeh, H. Abbaslou, A.T. Amlashi, P. Alidoust, Modeling of bentonite/sepiolite plastic concrete compressive strength using artificial neural network and support vector machine, Front. Struct. Civ. Eng. 13 (1) (Feb. 2019) 215–239, https://doi.org/10.1007/s11709-018-0489-z.
[86] A. Ashrafian, F. Shokri, M.J. Taheri Amiri, Z.M. Yaseen, M. Rezaie-Balf, Compressive strength of Foamed Cellular Lightweight Concrete simulation: new development of hybrid artificial intelligence model, Construct. Build. Mater. 230 (Jan. 2020), 117048, https://doi.org/10.1016/j.conbuildmat.2019.117048.
[87] A.T. Amlashi, P. Alidoust, A.R. Ghanizadeh, S. Khabiri, M. Pazhouhi, M.S. Monabati, Application of computational intelligence and statistical approaches for auto-estimating the compressive strength of plastic concrete, Eur. J. Environ. Civ. Eng. 26 (8) (Jun. 2022) 3459–3490, https://doi.org/10.1080/19648189.2020.1803144.
[88] H. Verbeeck, R. Samson, F. Verdonck, R. Lemeur, Parameter sensitivity and uncertainty of the forest carbon flux model FORUG: a Monte Carlo analysis, Tree Physiol. 26 (6) (Jun. 2006) 807–817, https://doi.org/10.1093/treephys/26.6.807.
[89] Y. Yang, Q. Zhang, A hierarchical analysis for rock engineering using artificial neural networks, Rock Mech. Rock Eng. 30 (4) (1997) 207–222, https://doi.org/10.1007/BF01045717.
[90] T. Shaikhina, N.A. Khovanova, Handling limited datasets with neural networks in medical applications: a small-data approach, Artif. Intell. Med. 75 (Jan. 2017) 51–63, https://doi.org/10.1016/j.artmed.2016.12.003.
[91] D.-C. Feng, et al., Machine learning-based compressive strength prediction for concrete: an adaptive boosting approach, Construct. Build. Mater. 230 (Jan. 2020), 117000, https://doi.org/10.1016/j.conbuildmat.2019.117000.
[92] B.A. Young, A. Hall, L. Pilon, P. Gupta, G. Sant, Can the compressive strength of concrete be estimated from knowledge of the mixture proportions?: new insights from statistical analysis and machine learning methods, Cement Concr. Res. 115 (Jan. 2019) 379–388, https://doi.org/10.1016/j.cemconres.2018.09.006.